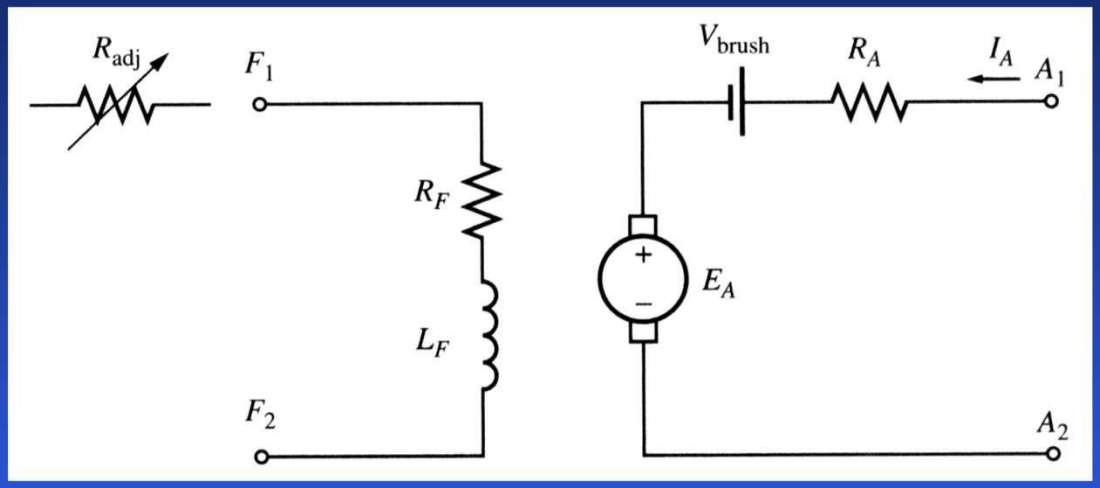
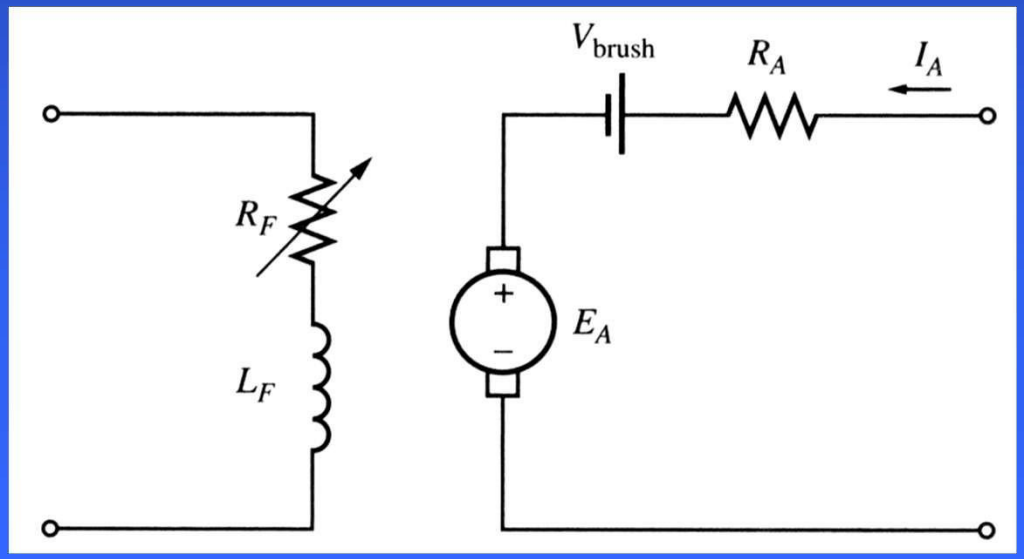
Equivalent circuit of a DC motor

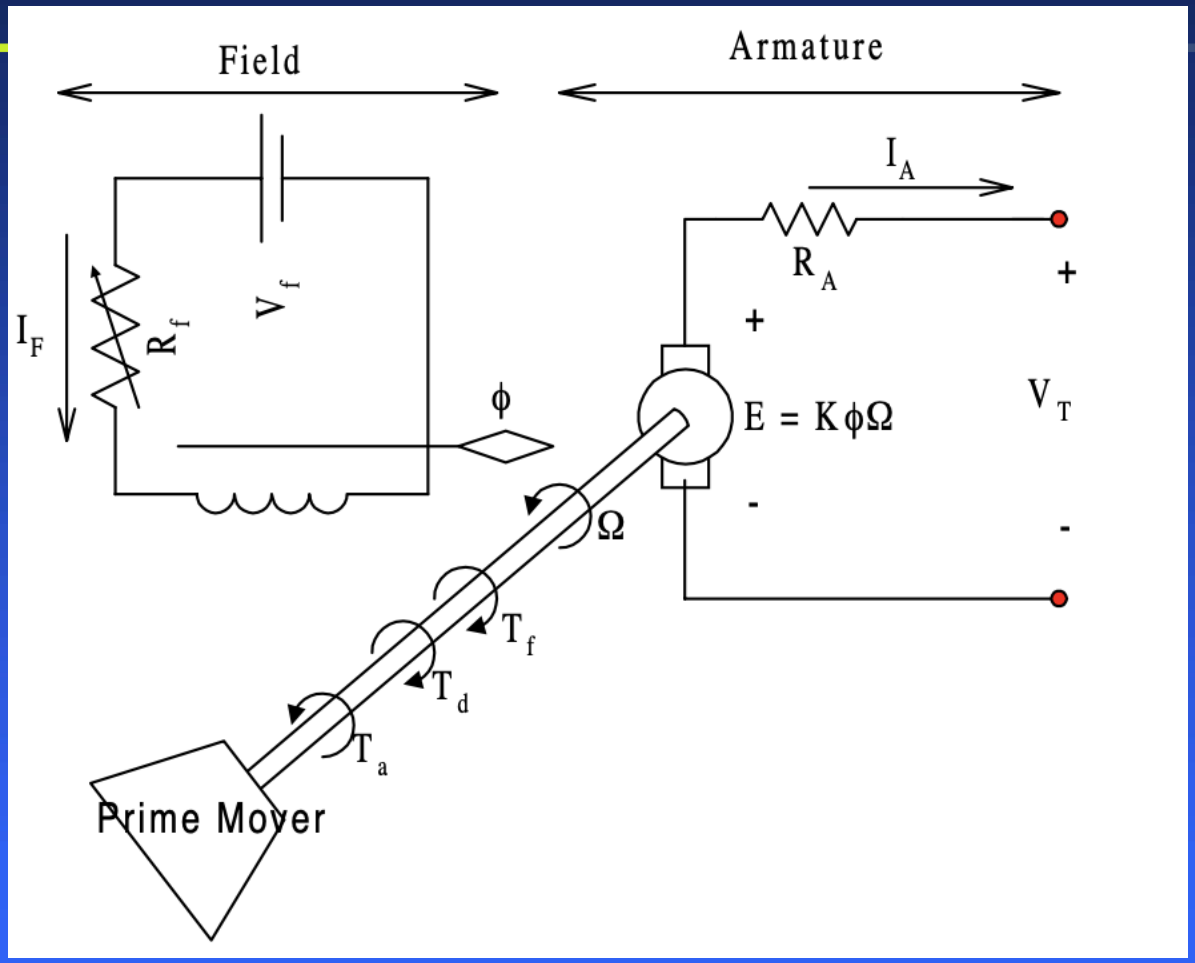
The armature circuit (the entire rotor structure) is represented by an ideal voltage source E_A and a resistor R_A. A battery V_{brush} in the opposite to a current flow in the machine direction indicates brush voltage drop.

The field coils producing the magnetic flux are represented by inductor L_F and resistor R_F. The resistor R_{adj} represents an external variable resistor (sometimes lumped together with the field coil resistance) used to control the amount of current in the field circuit.





Seperately Excited DC Generator



Item	*Linear Machine*	*Rotating Machine*
Velocity/Speed	u m/s	Ω rad/s or n rpm
Current, Armature	I amperes	I_A amperes
Current, Field	----------	I_f amperes
Magnetic Field	B Webers/m^2	ϕ Webers (per pole)
Voltage, Induced	$e = BLu$ newtons	$E = K\phi\Omega = k\phi n$ volts
Voltage, Terminal	v Volts	V_T Volts
Force/Torque (Developed)	$f_d = BLi$ newtons	$T_d = K\phi I_A$ N-m
Force/Torque (Applied)	f_a newtons	T_a newton-meters
Resistance, Armature	r ohms	R_A ohms
Mechanical Power Dev.	$f_d u$ watts	$T_d \Omega$ watts
Electrical Power Dev.	ei watts	EI_A watts
Power Equivalence	$ei = f_d u$	$EI_A = T_d \Omega$
Machine Constants*	----------	K,k

*The constant for a given machine lumps together all the effects of the *unchanging* quantities such as the number of poles, dimensions of the armature core, number of armature coils and the method of interconnecting them.

Rated capacity of a DC machine

The **rated capacity** of a DC machine (motor or generator) refers to the maximum continuous power output it can deliver under specific conditions without overheating or risking damage. This rating is determined by the machine's design, materials, and cooling system, and it includes several key parameters:

Rated Power (W or kW)
Rated Voltage (V)
Rated Current (A)
Rated Speed (rpm)
Rated Torque (Nm) (for motors)

Rated capacity of a DC machine

These ratings ensure that the DC machine operates efficiently and safely.

Operating within these limits helps maintain the longevity and reliability of the machine, while exceeding these values can result in decreased performance, increased wear, and potential damage.

FULL LOAD

In the context of a DC machine, **full load** refers to the condition when the machine is operating at its rated capacity. This typically means that the machine is delivering its maximum rated power output, which includes both the rated current and voltage, under specified conditions.

FULL LOAD

For a DC motor, full load means:

1. **Current**: The motor draws its rated current.

2. **Torque**: The motor delivers its rated torque.

3. **Speed**: The motor operates at its rated speed (though speed may decrease slightly under full load due to voltage drops in the armature resistance).

4. **Temperature**: The machine is within its thermal limit (i.e., it doesn't overheat over prolonged operation).

FULL LOAD

For a DC generator, full load indicates that the generator is supplying its rated current at its rated voltage to an external load, and it is operating within its designed temperature rise limits.

In both cases, operating beyond full load risks overheating, excessive wear, or damage to the machine.

Equivalent circuit of a DC motor

Sometimes, when the brush drop voltage is small, it may be left out. Also, some DC motors have more than one field coil…

The internal generated voltage in the machine is

$$E_A = K\phi\omega \tag{5.43.1}$$

The induced torque developed by the machine is

$$\tau_{ind} = K\phi I_A \tag{5.43.2}$$

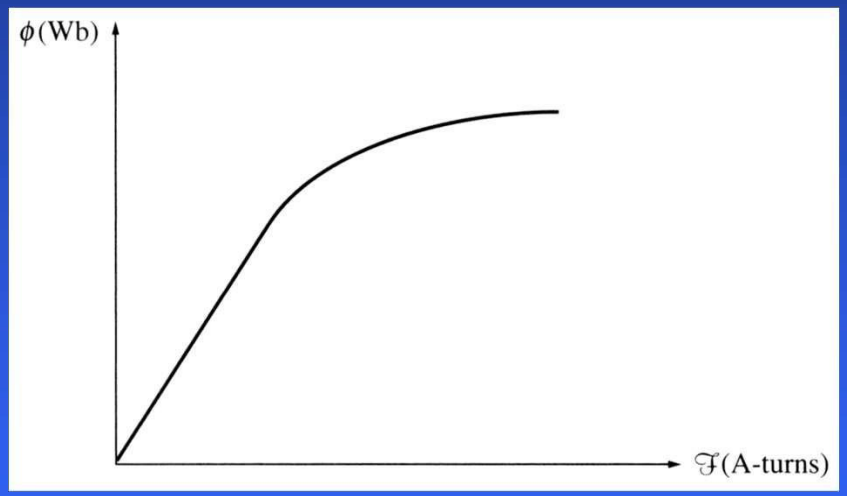
Here K is the constant depending on the design of a particular DC machine (number and commutation of rotor coils, etc.) and ϕ is the total flux inside the machine.

Note that for a single rotating loop $K = \pi/2$.

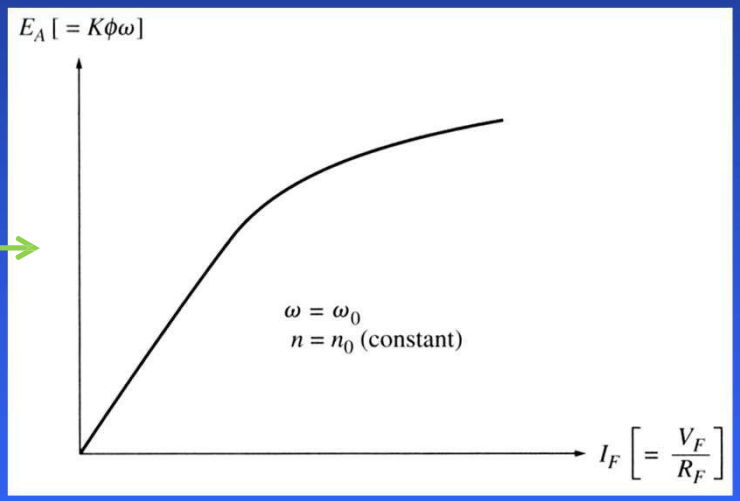
Magnetization curve of a DC machine

The internal generated voltage E_A is directly proportional to the flux in the machine and the speed of its rotation.

The field current in a DC machine produces a field mmf $\mathcal{F} = N_F I_F$, which produces a flux in the machine according to the magnetization curve.

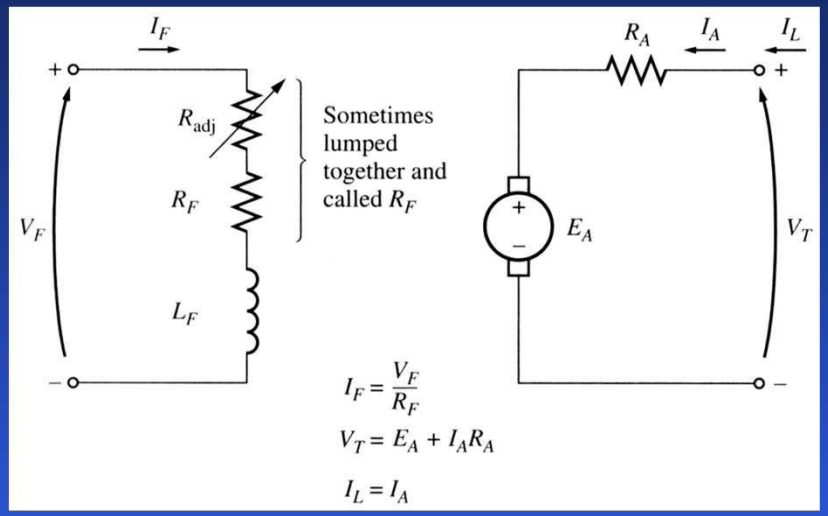


or in terms of internal voltage vs. field current for a given speed.

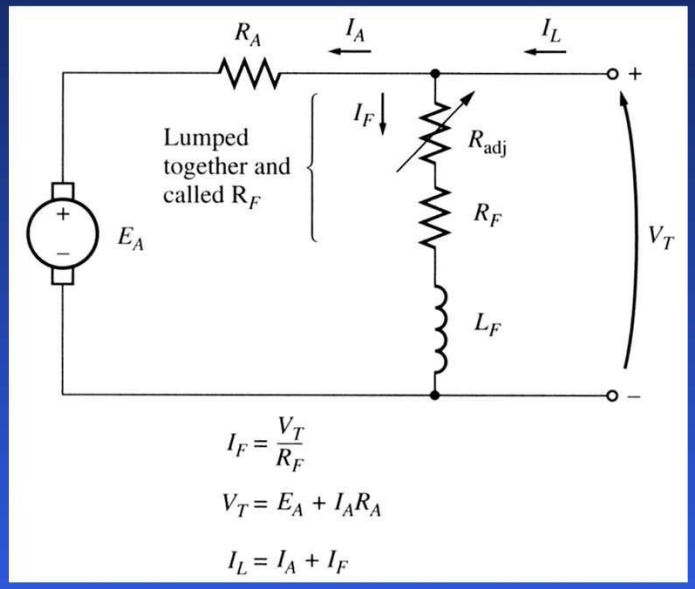


To get the maximum possible power per weight out of the machine, most motors and generators are operating near the saturation point on the magnetization curve. Therefore, when operating at full load, often a large increase in current I_F may be needed for small increases in the generated voltage E_A.

Motor types: Separately excited and Shunt DC motors



Note: when the voltage to the field circuit is assumed constant, there is no difference between them…



Separately excited DC motor:
a field circuit is supplied from a separate constant voltage power source.

Shunt DC motor:
a field circuit gets its power from the armature terminals of the motor.

For the armature circuit of these motors:

$$V_T = E_A + I_A R_A \qquad (5.45.1)$$

Shunt motor: terminal characteristic

A terminal characteristic of a machine is a plot of the machine's output quantities vs. each other.

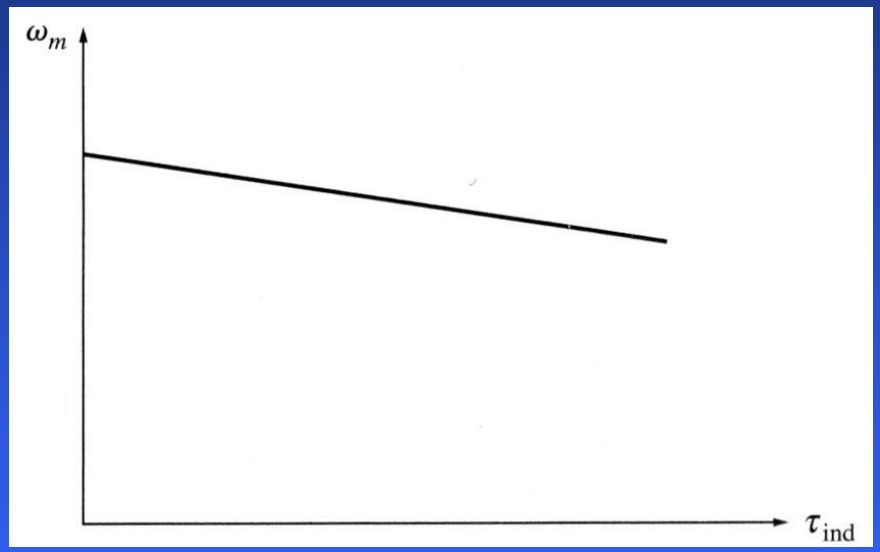
For a motor, the output quantities are shaft torque and speed. Therefore, the terminal characteristic of a motor is its output torque vs. speed.

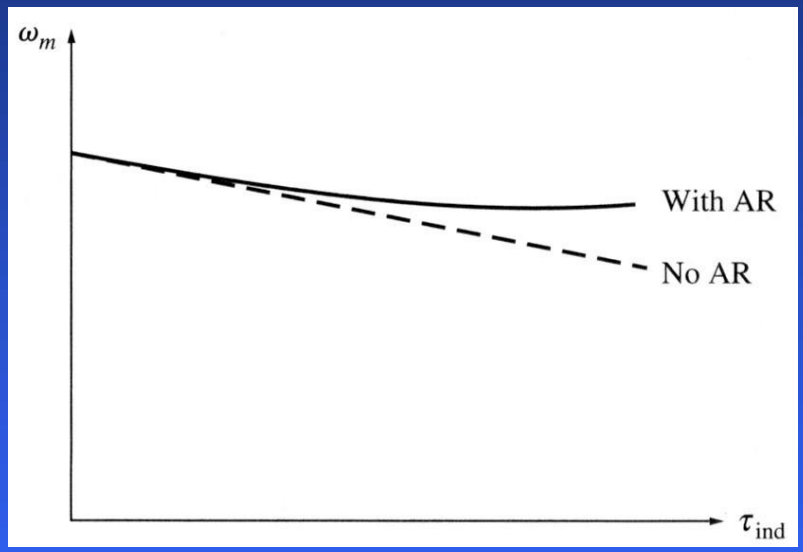
If the load on the shaft increases, the load torque τ_{load} will exceed the induced torque τ_{ind}, and the motor will slow down. Slowing down the motor will decrease its internal generated voltage (since $E_A = K\phi\omega$), so the armature current increases ($I_A = (V_T - E_A)/R_A$). As the armature current increases, the induced torque in the motor increases (since $\tau_{ind} = K\phi I_A$), and the induced torque will equal the load torque at a lower speed ω.

$$\omega = \frac{V_T}{K\phi} - \frac{R_A}{(K\phi)^2}\tau_{ind} \qquad (5.46.1)$$

Shunt motor: terminal characteristic

Assuming that the terminal voltage and other terms are constant, the motor's speed vary linearly with torque.



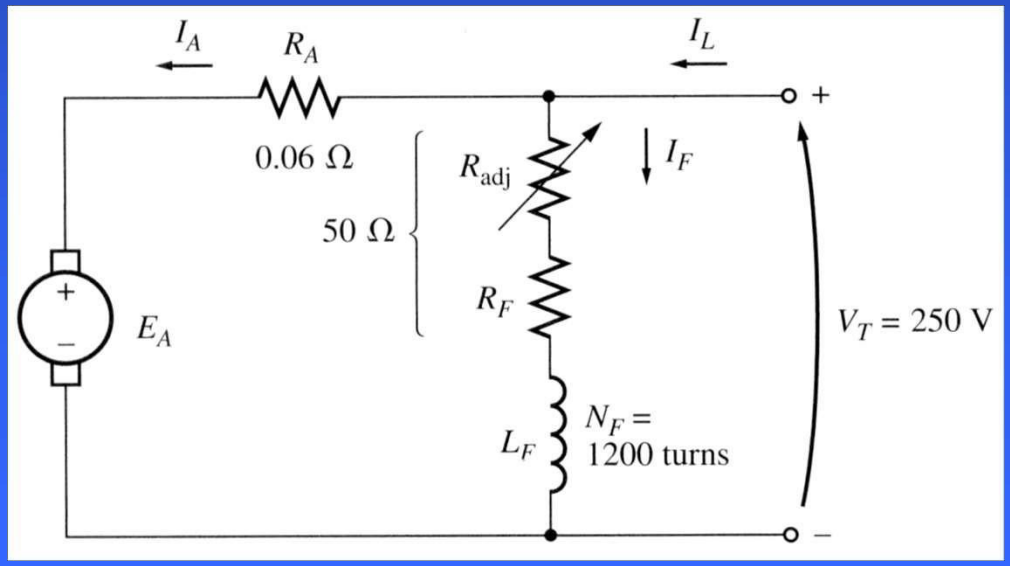


However, if a motor has an armature reaction, flux-weakening reduces the flux when torque increases. Therefore, the motor's speed will increase. If a shunt (or separately excited) motor has compensating windings, and the motor's speed and armature current are known for any value of load, it's possible to calculate the speed for any other value of load.

Shunt motor: terminal characteristic – Example

Example 5.1: A 50 hp, 250 V, 1200 rpm DC shunt motor **with** compensating windings has an armature resistance (including the brushes, compensating windings, and interpoles) of 0.06 Ω. Its field circuit has a total resistance $R_{adj} + R_F$ of 50 Ω, which produces a no-load speed of 1200 rpm. The shunt field winding has 1200 turns per pole.

a) Find the motor speed when its input current is 100 A.
b) Find the motor speed when its input current is 200 A.
c) Find the motor speed when its input current is 300 A.
d) Plot the motor torque-speed characteristic.



Shunt motor: terminal characteristic – Example

The internal generated voltage of a DC machine (with its speed expressed in rpm):

$$E_A = K\phi\omega$$

Since the field current is constant (both field resistance and V_T are constant) and since there are no armature reaction (due to compensating windings), we conclude that the flux in the motor is constant. The speed and the internal generated voltages at different loads are related as

$$\frac{E_{A2}}{E_{A1}} = \frac{K\phi\omega_2}{K\phi\omega_1} = \frac{n_2}{n_1}$$

Therefore:

$$n_2 = \frac{E_{A2}}{E_{A1}} n_1$$

At no load, the armature current is zero and therefore $E_{A1} = V_T = 250$ V.

Shunt motor: terminal characteristic – Example

a) Since the input current is 100 A, the armature current is

$$I_A = I_L - I_F = I_L - \frac{V_T}{R_F} = 100 - \frac{250}{50} = 95\ A$$

Therefore: $$E_A = V_T - I_A R_A = 250 - 95 \cdot 0.06 = 244.3 V$$

and the resulting motor speed is:

$$n_2 = \frac{E_{A2}}{E_{A1}} n_1 = \frac{244.3}{250} 1200 = 1173\ rpm$$

b) Similar computations for the input current of 200 A lead to n_2 = 1144 rpm.

c) Similar computations for the input current of 300 A lead to n_2 = 1115 rpm.

d) To plot the output characteristic of the motor, we need to find the torque corresponding to each speed. At no load, the torque is zero.





Shunt motor: terminal characteristic – Example

Since the induced torque at any load is related to the power converted in a DC motor:

$$P_{conv} = E_A I_A = \tau_{ind} \omega$$

the induced torque is $$\tau_{ind} = \frac{E_A I_A}{\omega}$$

For the input current of 100 A: $$\tau_{ind} = \frac{2443 \cdot 95}{2\pi \cdot 1173/60} = 190\ N\text{-}m$$

For the input current of 200 A: $$\tau_{ind} = \frac{2383 \cdot 195}{2\pi \cdot 1144/60} = 388\ N\text{-}m$$

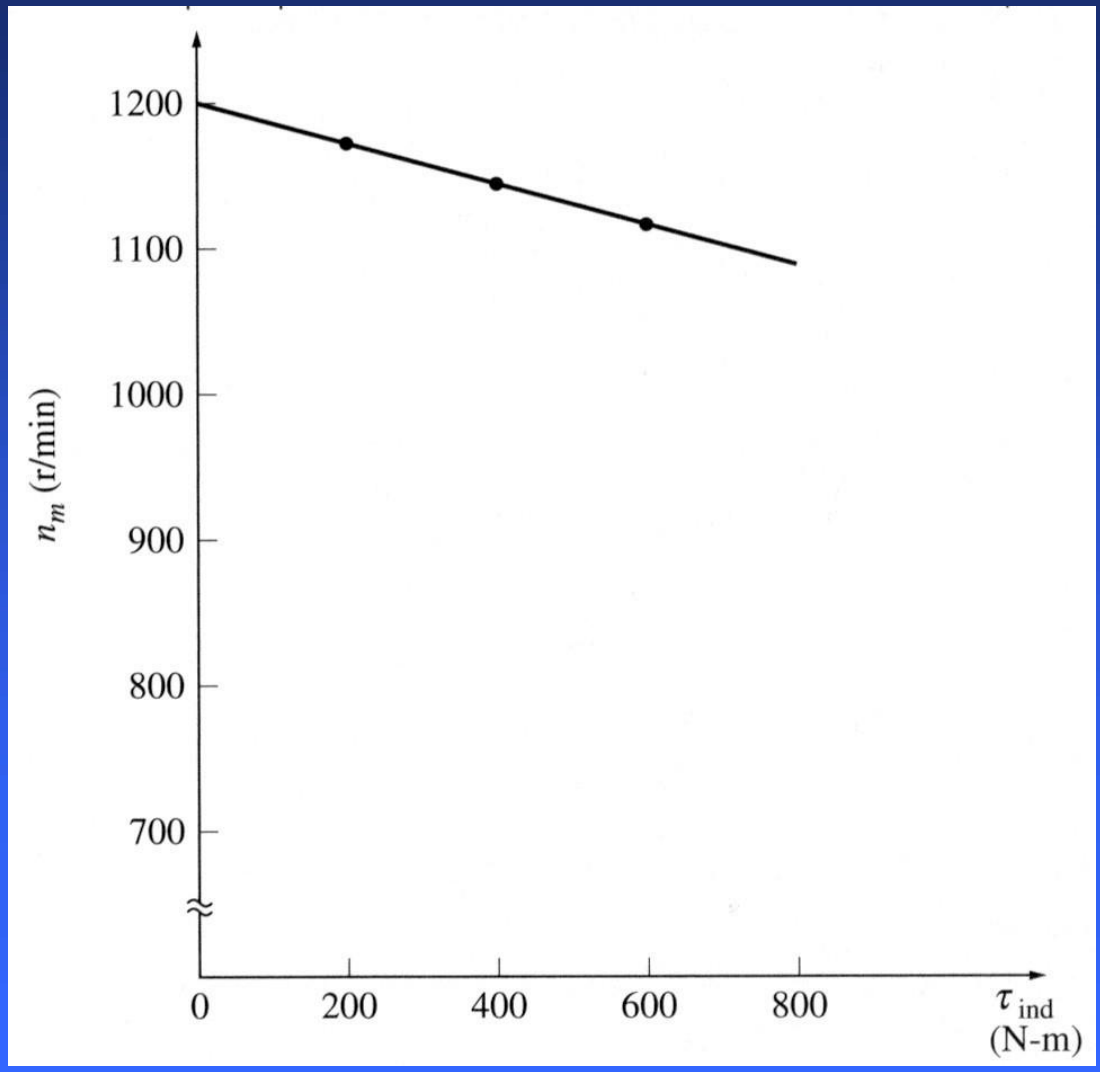
For the input current of 300 A: $$\tau_{ind} = \frac{2323 \cdot 295}{2\pi \cdot 1115/60} = 587\ N\text{-}m$$





Shunt motor: terminal characteristic – Example

The torque-speed characteristic of the motor is:



Shunt motor: Nonlinear analysis

The flux ϕ and therefore the internal generated voltage E_A of a DC machine are nonlinear functions of its mmf and must be determined based on the magnetization curve. Two main contributors to the mmf are its field current and the armature reaction (if present).

Since the magnetization curve is a plot of the generated voltage vs. field current, the effect of changing the field current can be determined directly from the magnetization curve.

If a machine has armature reaction, its flux will reduce with increase in load. The total mmf in this case will be

$$\mathsf{F}_{net} = N_F I_F - \mathsf{F}_{AR} \qquad (5.53.1)$$

It is customary to define an equivalent field current that would produce the same output voltage as the net (total) mmf in the machine:

$$I_F^* = I_F - \frac{\mathsf{F}_{AR}}{N_F} \qquad (5.53.2)$$

Shunt motor: Nonlinear analysis

Conducting a nonlinear analysis to determine the internal generated voltage in a DC motor, we may need to account for the fact that a motor can be running at a speed other than the rated one.

The voltage induced in a DC machine is

$$E_A = K\phi\omega \tag{5.54.1}$$

For a given effective field current, the flux in the machine is constant and the internal generated voltage is directly proportional to speed:

$$\frac{E_A}{E_{A0}} = \frac{n}{n_0} \tag{5.54.2}$$

Where E_{A0} and n_0 represent the reference (rated) values of voltage and speed, respectively. Therefore, if the reference conditions are known from the magnetization curve and the actual E_A is computed, the actual speed can be determined.

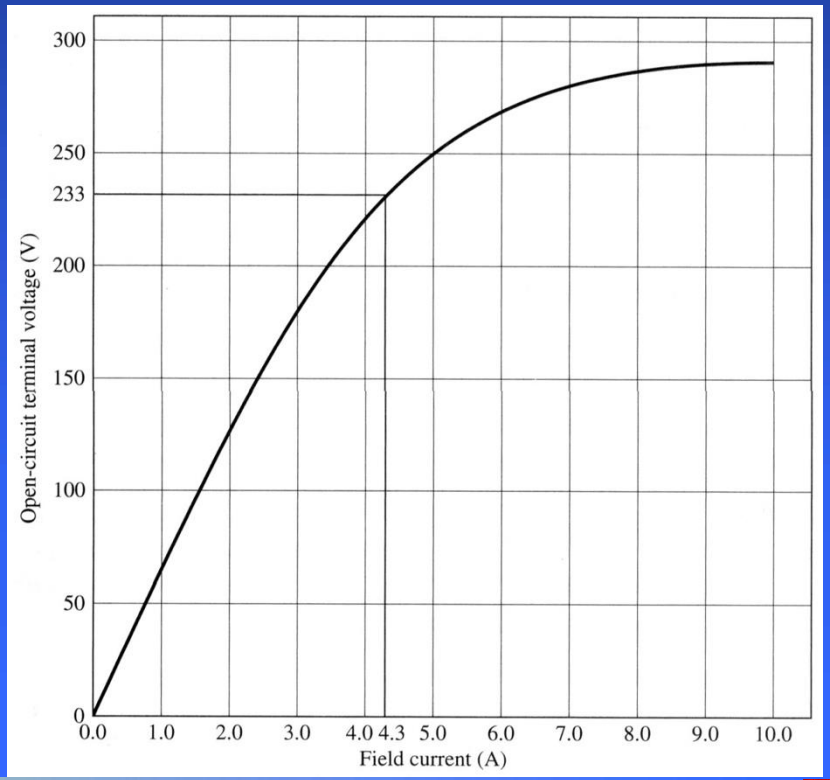




Shunt motor: Nonlinear analysis – Example

Example 5.2: A 50 hp, 250 V, 1200 rpm DC shunt motor **without** compensating windings has an armature resistance (including the brushes and interpoles) of 0.06 Ω. Its field circuit has a total resistance $R_{adj} + R_F$ of 50 Ω, which produces a no-load speed of 1200 rpm. The shunt field winding has 1200 turns per pole. The armature reaction produces a demagnetizing mmf of 840 A-turns at a load current of 200A. The magnetization curve is shown.

a) Find the motor speed when its input current is 200 A.
b) How does the motor speed compare to the speed of the motor from Example 5.1 (same motor but **with** compensating windings) with an input current of 200 A?
c) Plot the motor torque-speed characteristic.



Shunt motor: Nonlinear analysis – Example

a) Since the input current is 200 A, the armature current is

$$I_A = I_L - I_F = I_L - \frac{V_T}{R_F} = 200 - \frac{250}{50} = 195\ A$$

Therefore: $$E_A = V_T - I_A R_A = 250 - 195 \cdot 0.06 = 238.3\ V$$

At the given current, the demagnetizing mmf due to armature reaction is 840 A-turns, so the effective shunt field current of the motor is

$$I_F^* = I_F - \frac{\mathsf{F}_{AR}}{N_F} = 5 - \frac{840}{1200} = 4.3\ A$$

From the magnetization curve, this effective field current will produce an internal voltage of E_{A0} = 233 V at a speed of 1200 rpm. For the actual voltage, the speed is

$$n = \frac{E_A}{E_{A0}} n_0 = \frac{238.3}{233} 1200 = 1227\ rpm$$

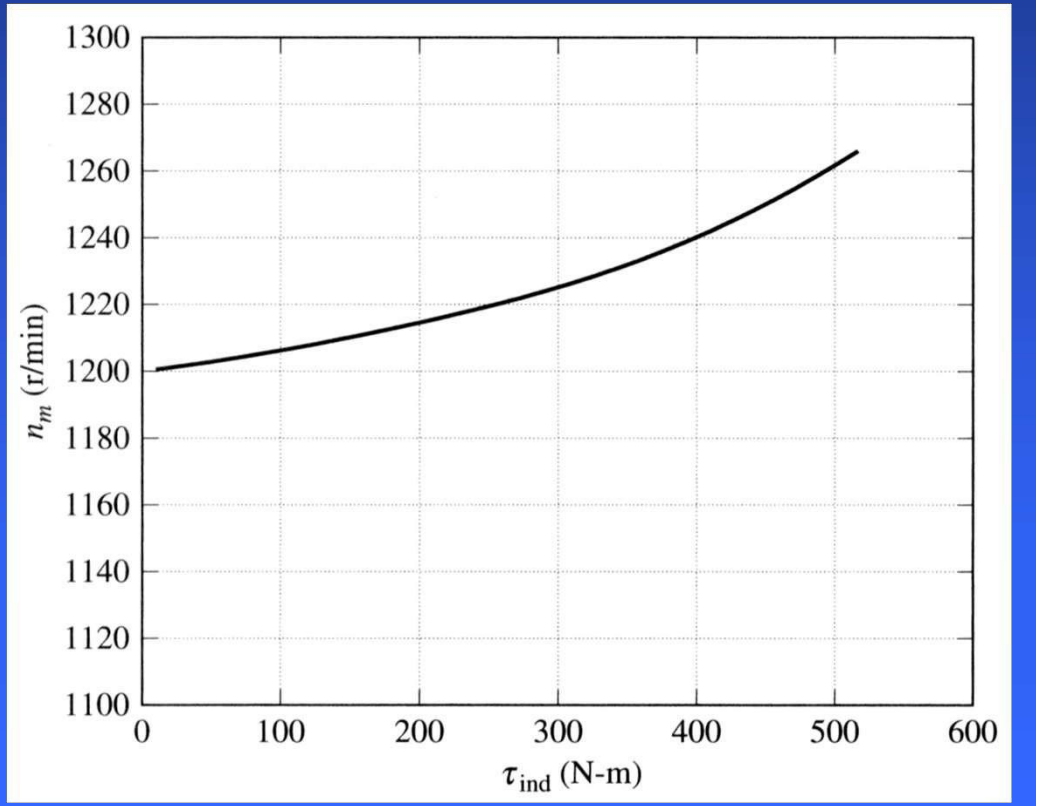




Shunt motor: Nonlinear analysis – Example

b) A speed of a motor with compensating windings was 1144 rpm when the input current was 200 A. We notice that the speed of the motor with armature reactance is higher than the speed of the motor without armature reactance. This increase is due to the flux weakening.

c) Assuming that the mmf due to the armature reaction varies linearly with the increase in current, and repeating the same computations for many different load currents, the motor's torque-speed characteristic can be plotted.



Shunt motor: Speed control

There are two methods to control the speed of a shunt DC motor:

1. Adjusting the field resistance R_F (and thus the field flux)
2. Adjusting the terminal voltage applied to the armatüre (rotor winding)

1. Adjusting the field resistance

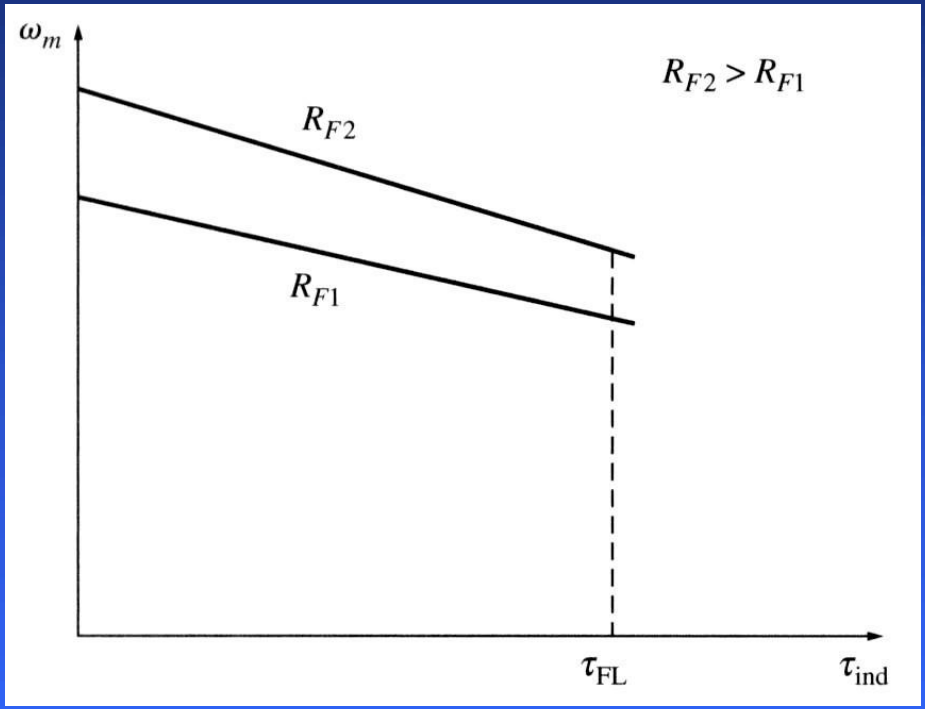
1) Increasing field resistance R_F decreases the field current ($I_F = V_T/R_F$);
2) Decreasing field current I_F decreases the flux ϕ;
3) Decreasing flux decreases the internal generated voltage ($E_A = K\phi\omega$);
4) Decreasing E_A increases the armature current ($I_A = (V_T - E_A)/R_A$);
5) Changes in armature current dominate over changes in flux; therefore, increasing I_A increases the induced torque ($\tau_{ind} = K\phi I_A$);
6) Increased induced torque is now larger than the load torque τ_{load} and, therefore, the speed ω increases;
7) Increasing speed increases the internal generated voltage E_A;
8) Increasing E_A decreases the armature current I_A…
9) Decreasing I_A decreases the induced torque until $\tau_{ind} = \tau_{load}$ at a higher speed ω.




Shunt motor: Speed control

The effect of increasing the field resistance within a normal load range: from no load to full load.

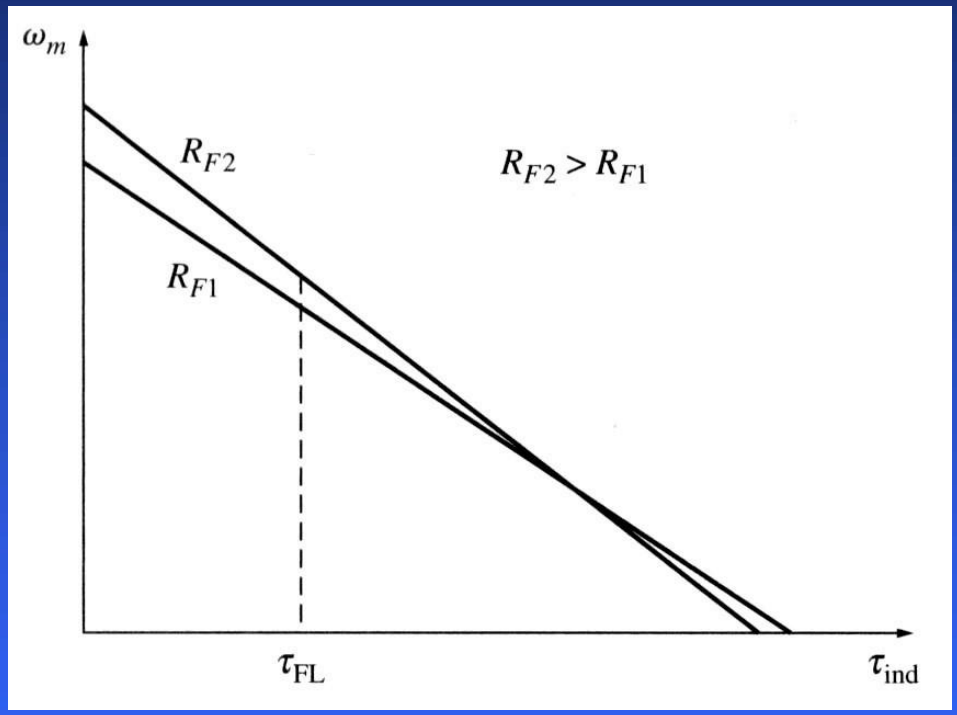
Increase in the field resistance increases the motor speed. Observe also that the slope of the speed-torque curve becomes steeper when field resistance increases.



Shunt motor: Speed control

The effect of increasing the field resistance with over an entire load range: from no-load to stall.

At very slow speeds (overloaded motor), an increase in the field resistance decreases the speed (compare R_{F1} with R_{F2}). In this region, the increase in armature current is no longer large enough to compensate for the decrease in flux.



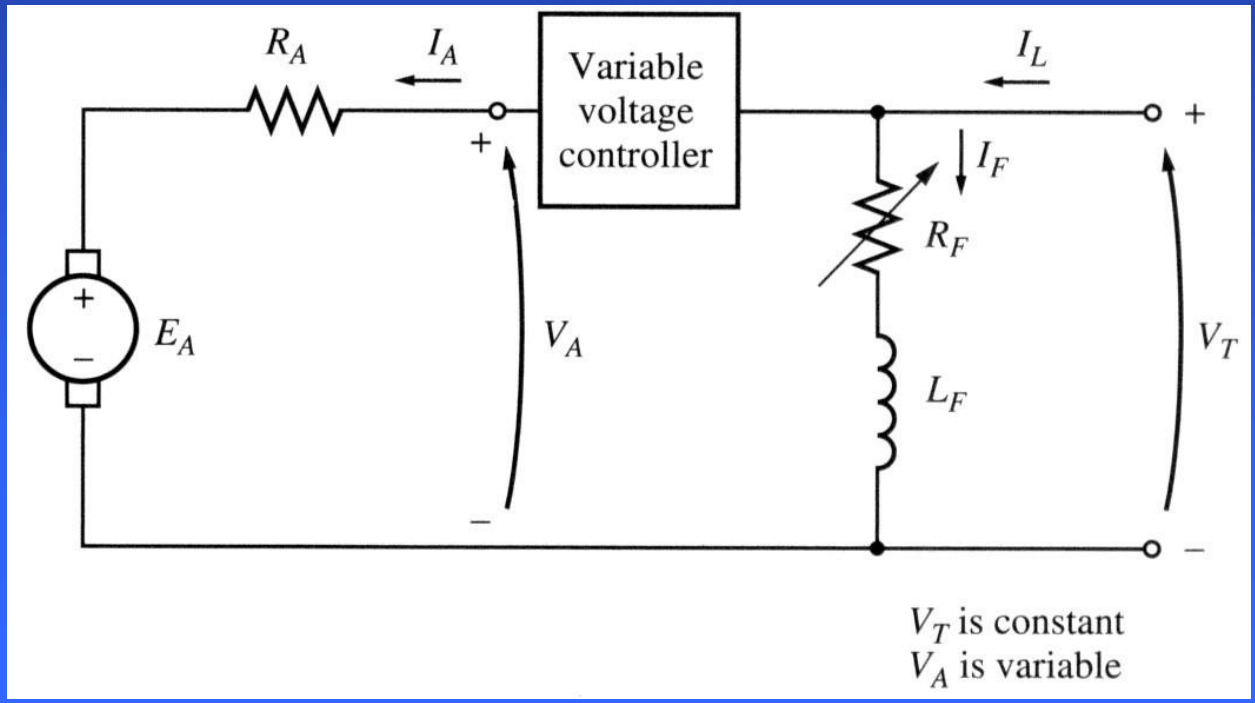
Some small DC motors used in control circuits may operate at speeds close to stall conditions. For such motors, an increase in field resistance may have no effect (or opposite to the expected effect) on the motor speed. The result of speed control by field resistance is not predictable and, thus, this type of control is not very common.

Shunt motor: Speed control

2. Changing the armature voltage

This method implies changing the voltage applied to the armature of the motor without changing the voltage applied to its field. Therefore, the motor must be separately excited to use armature voltage control.

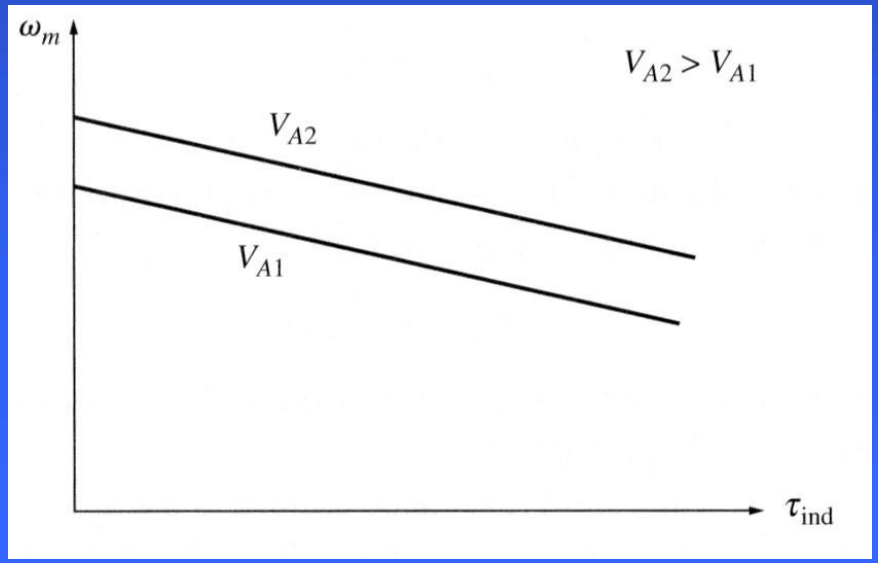
Armature voltage speed control



Shunt motor: Speed control

1) Increasing the armature voltage V_A increases the armature current ($I_A = (V_A - E_A)/R_A$);
2) Increasing armature current I_A increases the induced torque τ_{ind} ($\tau_{ind} = K\phi I_A$);
3) Increased induced torque τ_{ind} is now larger than the load torque τ_{load} and, therefore, the speed ω;
4) Increasing speed increases the internal generated voltage ($E_A = K\phi\omega$);
5) Increasing E_A decreases the armature current I_A…
6) Decreasing I_A decreases the induced torque until $\tau_{ind} = \tau_{load}$ at a higher speed ω.

Increasing the armature voltage of a separately excited DC motor does not change the slope of its torque-speed characteristic.



Shunt motor: Speed control

If a motor is operated at its rated terminal voltage, power, and field current, it will be running at the rated speed also called a base speed.

Field resistance control can be used for speeds above the base speed but not below it. Trying to achieve speeds slower than the base speed by the field circuit control, requires large field currents that may damage the field winding.

Since the armature voltage is limited to its rated value, no speeds exceeding the base speed can be achieved safely while using the armature voltage control.

Therefore, armature voltage control can be used to achieve speeds below the base speed, while the field resistance control can be used to achieve speeds above the base speed.

Shunt and separately excited DC motors have excellent speed control characteristic.

Shunt motor: Speed control

For the **armature voltage control**, the flux in the motor is constant. Therefore, the maximum torque in the motor will be constant too regardless the motor speed:

$$\tau_{\max} = K\phi I_{A,\max} \tag{5.64.1}$$

Since the maximum power of the motor is

$$P_{\max} = \tau_{\max}\omega \tag{5.64.2}$$

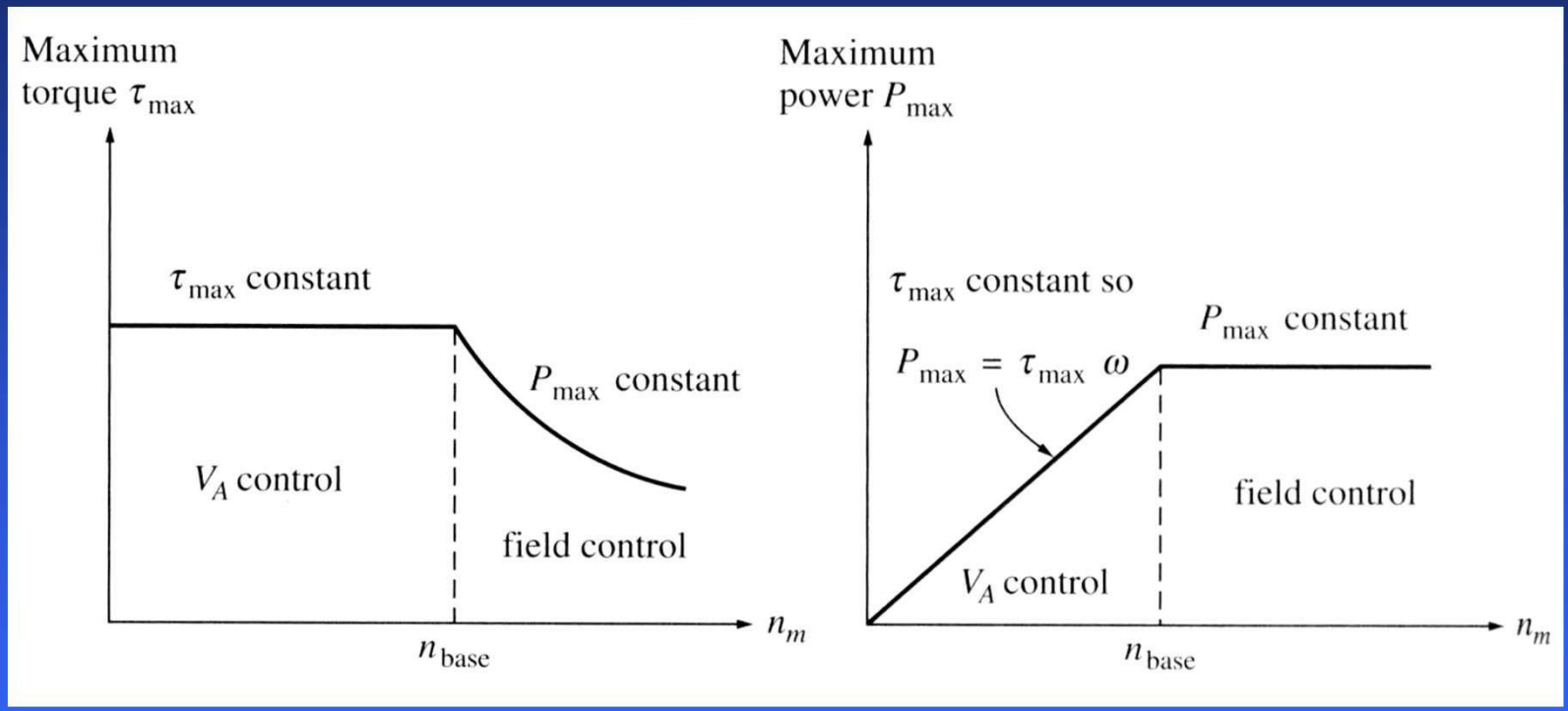
The maximum power out of the motor is directly proportional to its speed.

For the **field resistance control**, the maximum power out of a DC motor is constant, while the maximum torque is reciprocal to the motor speed.

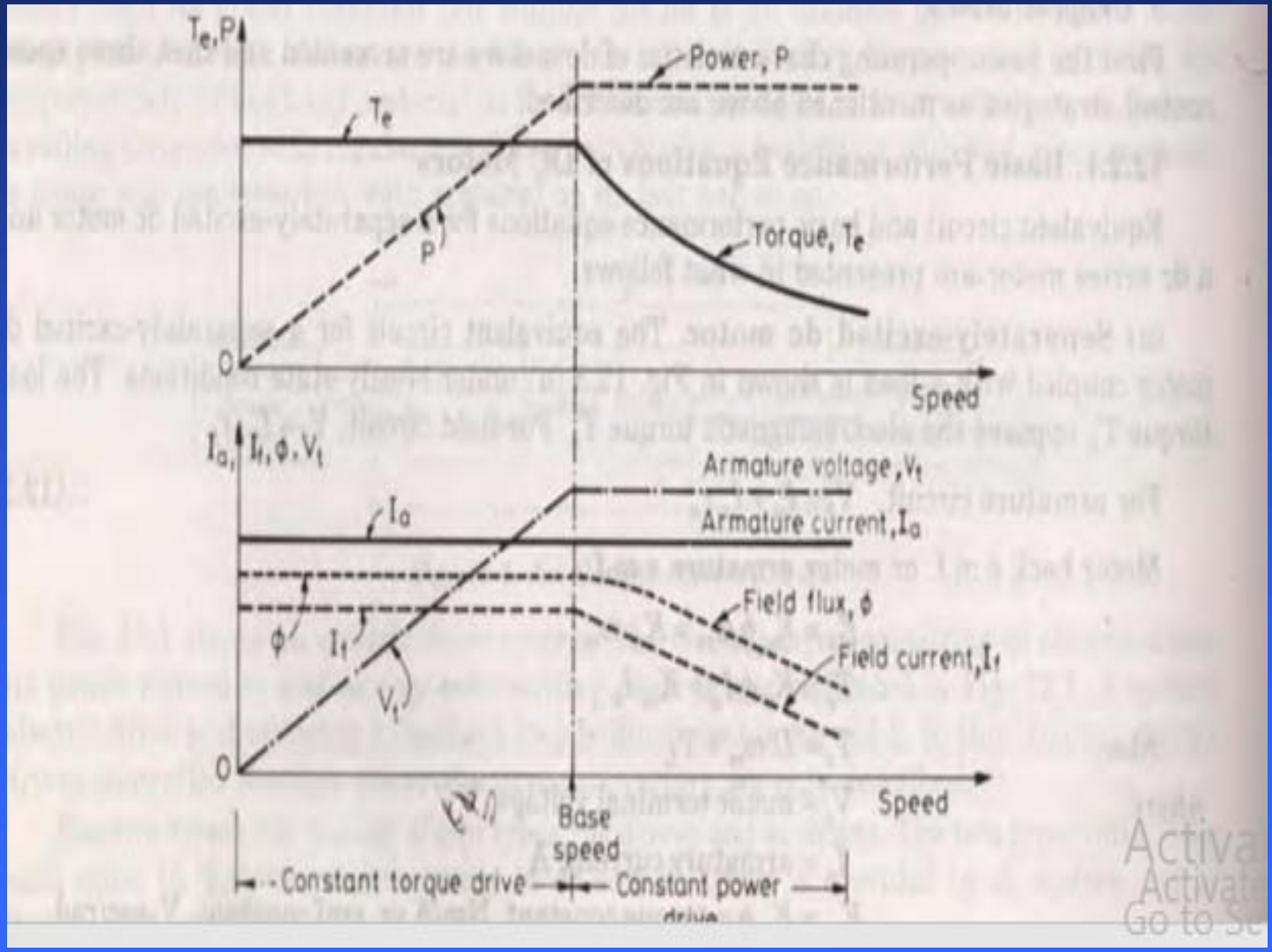




Shunt motor: Speed control



Torque and power limits as functions of motor speed for a shunt (or separately excited) DC motor.


T_e, P
Power, P
T_e
P
Torque, T_e
0
Speed
I_a, I_f, ϕ, V_t
Armature voltage, V_t
I_a
Armature current, I_a
Field flux, ϕ
ϕ
I_f
Field current, I_f
V_t
0
Speed
Base speed
Constant torque drive
Constant power drive

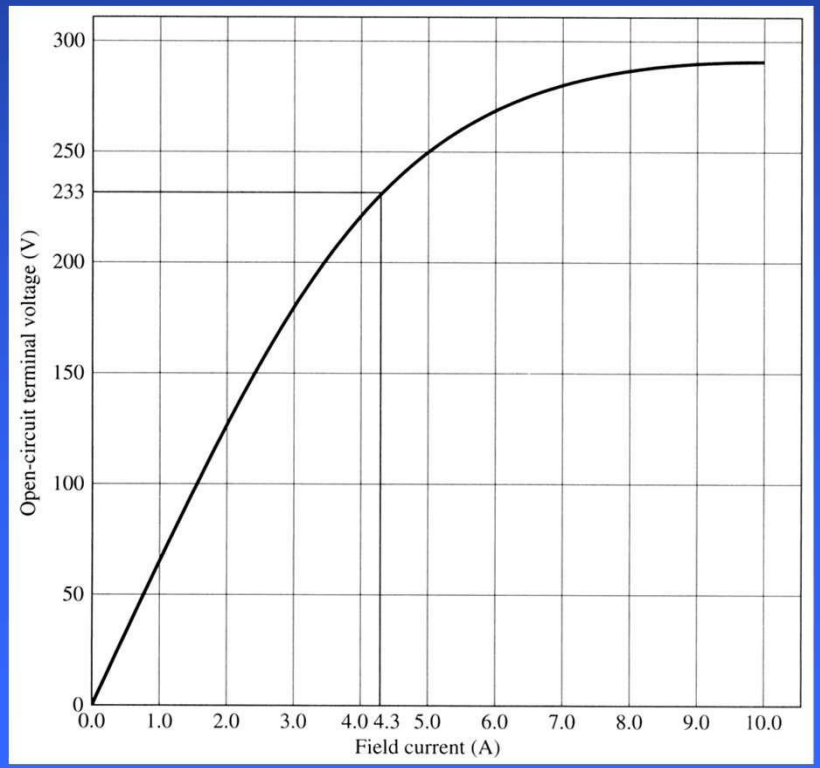
Armature voltage control : Armature voltage (V_t) is varied to obtain the speed below base speed. The armature current and field flux (Φ) are kept constant at their rated values. From the torque equation, it is seen that when I_A and Φ are constant the torque remains constant at its rated value to meet the required demand. So, this armature voltage control method is known as

constant-torque drive method The armature voltage (V_t) is varied from its zero voltage to rated voltage the power $P = V_T \cdot I_A$ increases from its zero value to rated power.

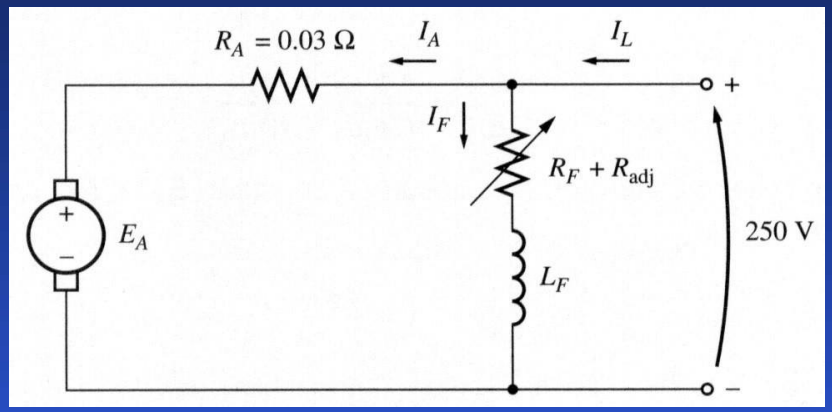
Shunt motor: Speed control: Ex

Example 5.3: A 100 hp, 250 V, 1200 rpm DC shunt motor with an armature resistance of 0.03 Ω and a field resistance of 41.67 Ω. The motor has compensating windings, so armature reactance can be ignored. Mechanical and core losses may be ignored also. The motor is driving a load with a line current of 126 A and an initial speed of 1103 rpm. Assuming that the armature current is constant and the magnetization curve is

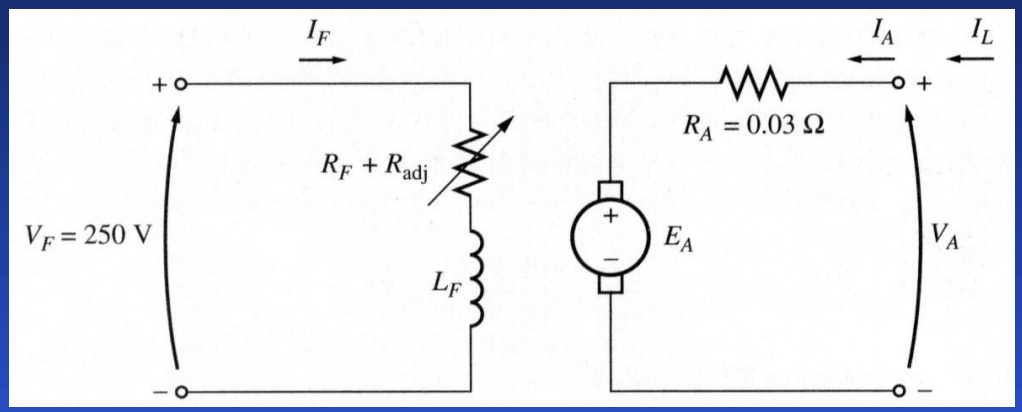
a) What is the motor speed if the field resistance is increased to 50 Ω?
b) Calculate the motor speed as a function of the field resistance, assuming a constant-current load.
c) Assuming that the motor next is connected as a separately excited and is initially running with V_A = 250 V, I_A = 120 A and at n = 1103 rpm while supplying a constant-torque load, estimate the motor speed if V_A is reduced to 200 V.



Shunt motor: Speed control: Ex



shunt



separately-excited

For the given initial line current of 126 A, the initial armature current will be

$$I_{A1} = I_{L1} - I_{F1} = 126 - \frac{250}{41.67} = 120\ A$$

Therefore, the initial generated voltage for the shunt motor will be

$$E_{A1} = V_T - I_{A1}R_A = 250 - 120 \cdot 0.03 = 246.4\ V$$

Shunt motor: Speed control: Ex

After the field resistance is increased to 50 Ω, the new field current will be

$$I_{F2} = \frac{250}{50} = 5\ A$$

The ratio of the two internal generated voltages is

$$\frac{E_{A2}}{E_{A1}} = \frac{K\phi_2\omega_2}{K\phi_1\omega_1} = \frac{\phi_2 n_2}{\phi_1 n_1}$$

Since the armature current is assumed constant, $E_{A1} = E_{A2}$ and, therefore

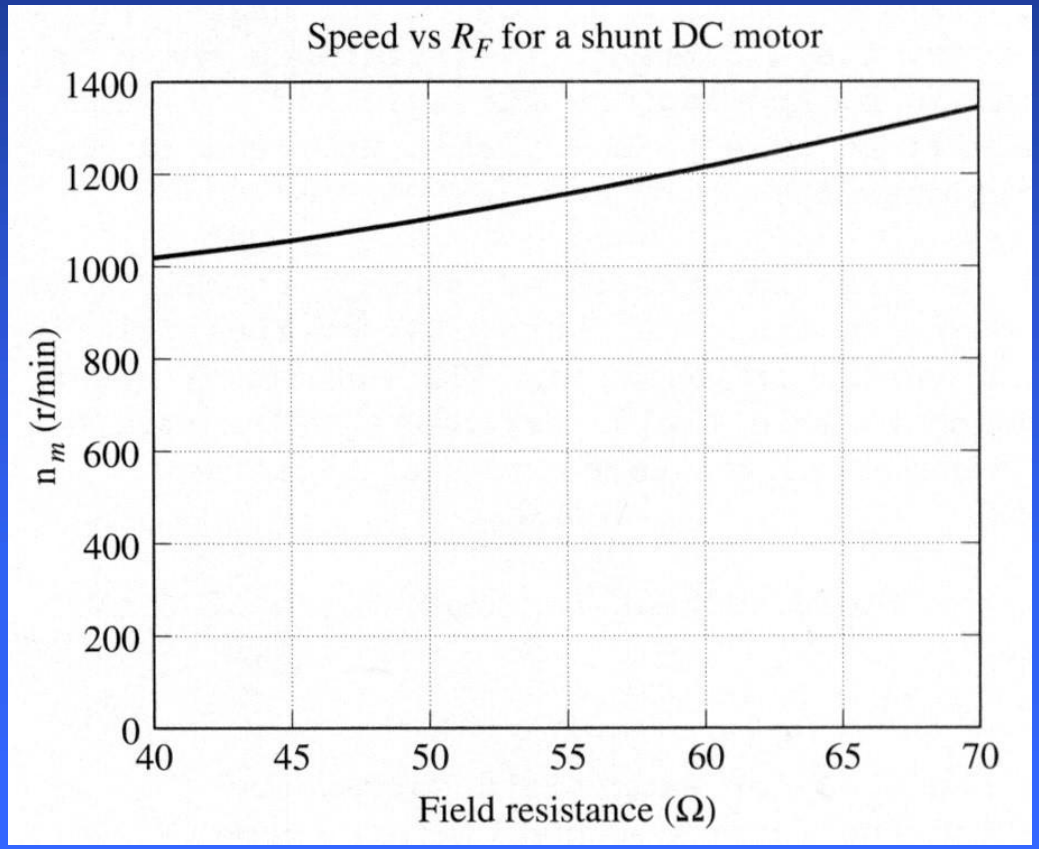
$$n_2 = \frac{\phi_1 n_1}{\phi_2}$$

The values of E_A on the magnetization curve are directly proportional to the flux. Therefore, the ratio of internal generated voltages equals to the ratio of the fluxes within the machine. From the magnetization curve, at I_F = 5A, E_{A1} = 250V, and at I_F = 6A, E_{A1} = 268V. Thus:

Shunt motor: Speed control: Ex

$$n_2 = \frac{\phi_1 n_1}{\phi_2} = \frac{E_{A1} n_1}{E_{A2}} = \frac{268}{250} 1103 = 1187 \; rpm$$

b) A speed vs. R_F characteristic is shown:



Shunt motor: Speed control: Ex

c) For a separately excited motor, the initial generated voltage is

$$E_{A1} = V_{T1} - I_{A1}R_A$$

Since

$$\frac{E_{A2}}{E_{A1}} = \frac{K\phi_2\omega_2}{K\phi_1\omega_1} = \frac{\phi_2 n_2}{\phi_1 n_1}$$

and since the flux ϕ is constant

$$n_2 = \frac{E_{A2}n_1}{E_{A1}}$$

Since the both the torque and the flux are constants, the armature current I_A is also constant. Then

$$n_2 = \frac{V_{T2} - I_{A2}R_A}{V_{T1} - I_{A1}R_A} n_1 = \frac{200 - 120 \cdot 0.03}{250 - 120 \cdot 0.03} 1103 = 879\ rpm$$





Shunt motor: The effect of an open field circuit

If the field circuit is left open on a shunt motor, its field resistance will be infinite. Infinite field resistance will cause a drastic flux drop and, therefore, a drastic drop in the generated voltage. The armature current will be increased enormously increasing the motor speed.

A similar effect can be caused by armature reaction. If the armature reaction is severe enough, an increase in load can weaken the flux causing increasing the motor speed. However, most loads have torque-speed curves whose torque *increases* with speed, so the increased speed of the motor increases its load, which increases its armature reaction, weakening its flux again. The weaker flux causes a further increase in speed, further increasing load, etc., until the motor overspeeds. This condition is known as *runaway*.

$$\tau_{ind} = K \downarrow\phi \Uparrow I_A$$

$$\omega = \frac{V_T}{K\phi} - \frac{R_A}{(K\phi)^2}\tau_{ind}$$

Motor types: The permanent-magnet DC motor

A permanent magnet DC (PMDC) motor is a motor whose poles are made out of permanent magnets.

Advantages:

1. Since no external field circuit is needed, there are no field circuit copper losses;
2. Since no field windings are needed, these motors can be considerable smaller.

Disadvantages:

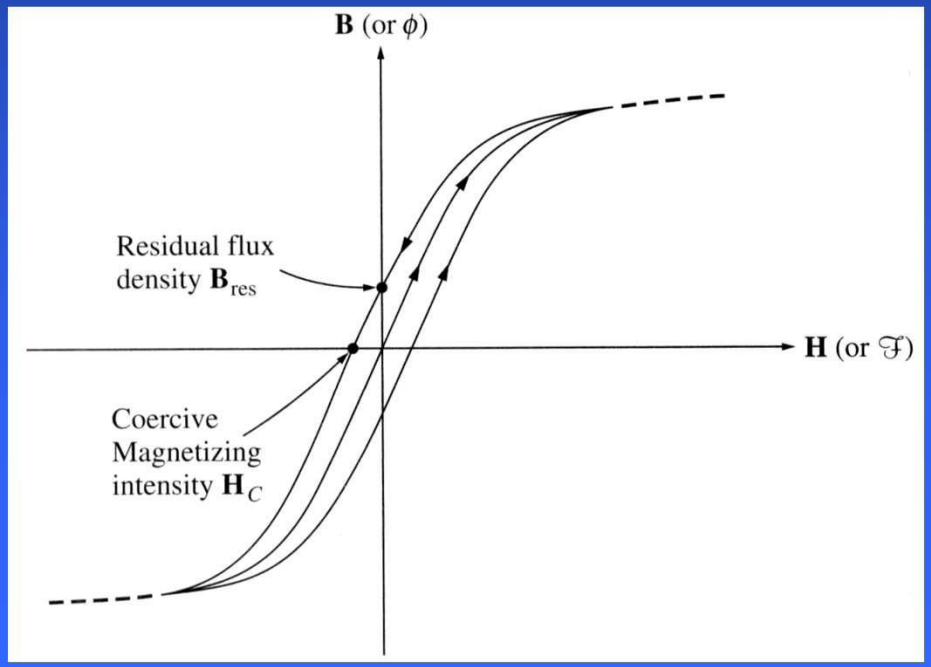
1. Since permanent magnets produces weaker flux densities then externally supported shunt fields, such motors have lower induced torque.
2. There is always a risk of demagnetization from extensive heating or from armature reaction effects (via armature mmf).

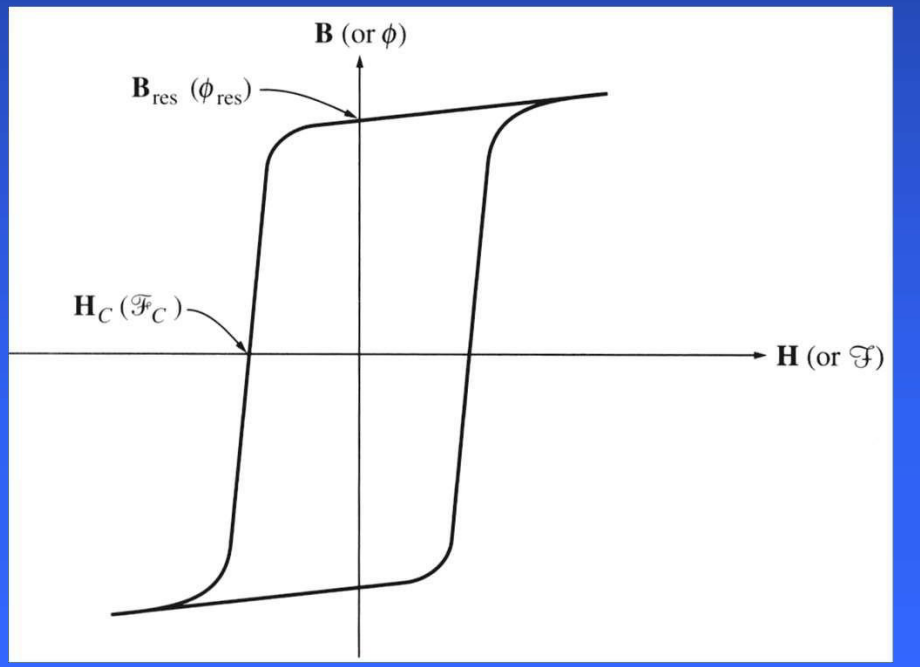


Motor types: The permanent-magnet DC motor

Normally (for cores), a ferromagnetic material is selected with small residual flux B_{res} and small coercive magnetizing intensity H_C.

However, a maximally large residual flux B_{res} and large coercive magnetizing intensity H_C are desirable for permanent magnets forming the poles of PMDC motors…

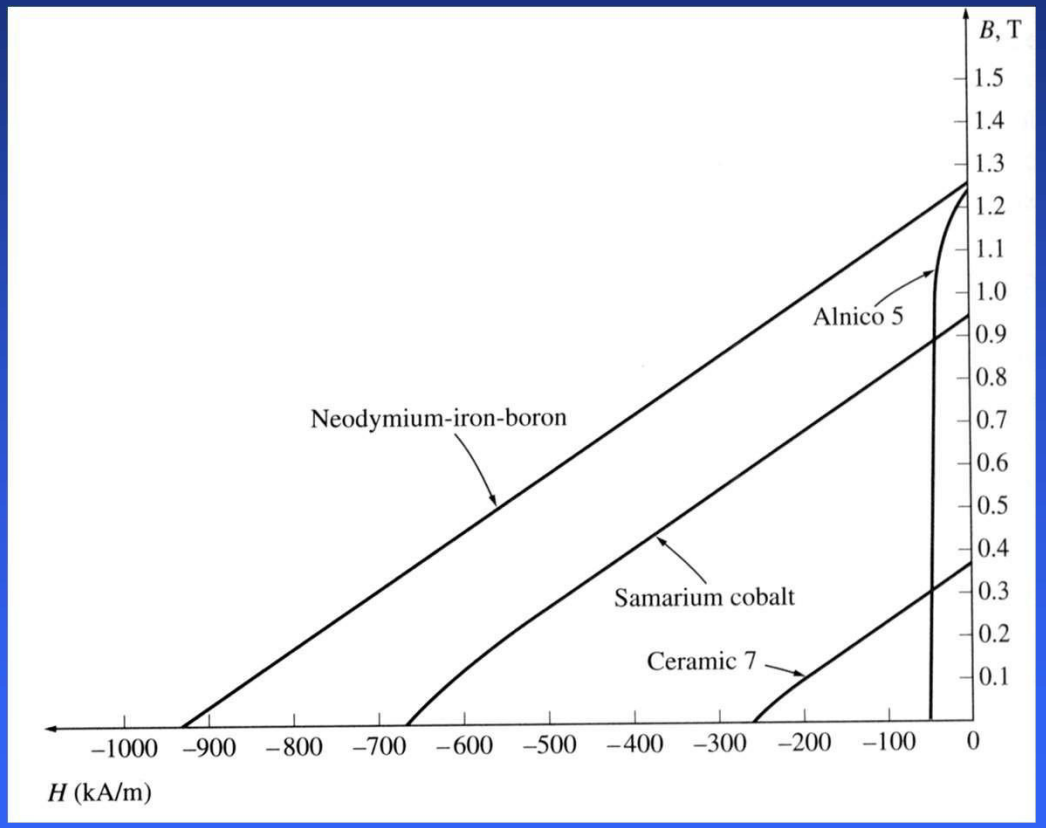




Motor types: The permanent-magnet DC motor

A comparison of magnetization curves of newly developed permanent magnets with that of a conventional ferromagnetic alloy (Alnico 5) shows that magnets made of such materials can produce the same residual flux as the best ferromagnetic cores.

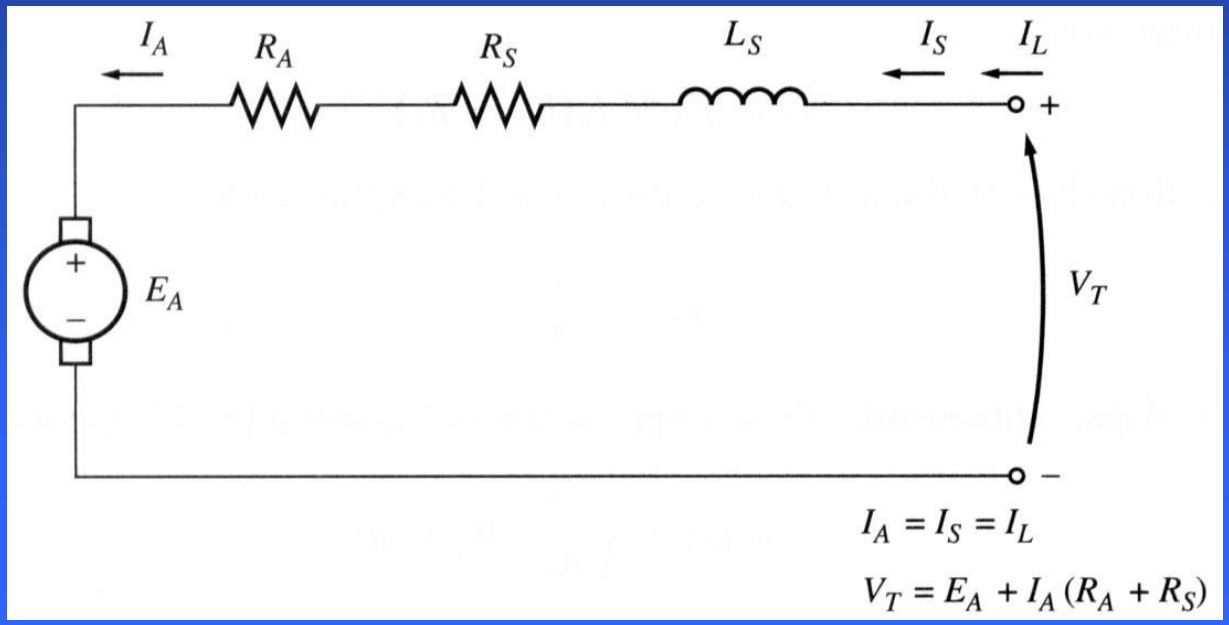
Design of permanent-magnet DC motors is quite similar to the design of shunt motors, except that the flux of a PMDC motor is fixed. Therefore, the only method of speed control available for PMDC motors is armature voltage control.



Motor types: The series DC motor

A series DC motor is a DC motor whose field windings consists of a relatively few turns connected in series with armature circuit. Therefore:

$$V_T = E_A + I_A(R_A + R_s) \tag{5.75.1}$$



The series winding is selected from thick wire to prevent voltage drop and burnout of the winding. Therefore, the number of turns is small.

Series motor: induced torque

The terminal characteristic of a series DC motor is quite different from that of the shunt motor since the flux is directly proportional to the armature current (assuming no saturation). An increase in motor flux causes a decrease in its speed; therefore, a series motor has a dropping torque-speed characteristic.

The induced torque in a series machine is

$$\tau_{ind} = K\phi I_A \tag{5.76.1}$$

Since the flux is proportional to the armature current:

$$\phi = cI_A \tag{5.76.2}$$

where c is a proportionality constant. Therefore, the torque is

$$\tau_{ind} = KcI_A^2 \tag{5.76.3}$$

Torque in the motor is proportional to the square of its armature current. Series motors supply the highest torque among the DC motors. Therefore, they are used as car starter motors, elevator motors etc.

Series motor: terminal characteristic

Assuming first that the magnetization curve is linear and no saturation occurs, flux is proportional to the armature current:

$$\phi = cI_A \tag{5.77.1}$$

Since the armature current is

$$I_A = \sqrt{\frac{\tau_{ind}}{Kc}} \tag{5.77.2}$$

and the armature voltage

$$E_A = K\phi\omega \tag{5.77.3}$$

The Kirchhoff's voltage law would be

$$V_T = E_A + I_A(R_A + R_S) = K\phi\omega + \sqrt{\frac{\tau_{ind}}{Kc}}\left(R_A + R_S\right) \tag{5.77.4}$$

Since (5.77.1), the torque:

$$\tau_{ind} = KcI_A^2 = \frac{K}{c}\phi^2 \tag{5.77.5}$$





Series motor: terminal characteristic

Therefore, the flux in the motor is

$$\phi = \sqrt{\frac{c}{K}}\sqrt{\tau_{ind}} \tag{5.78.1}$$

The voltage equation (5.77.4) then becomes

$$V_T = K\sqrt{\frac{c}{K}}\sqrt{\tau_{ind}}\,\omega + \sqrt{\frac{\tau_{ind}}{Kc}}\left(R_A + R_S\right) \tag{5.78.2}$$

which can be solved for the speed:

$$\omega = \frac{V_T}{\sqrt{Kc}}\frac{1}{\sqrt{\tau_{ind}}} - \frac{R_A + R_S}{Kc} \tag{5.78.3}$$

The speed of unsaturated series motor inversely proportional to the square root of its torque.

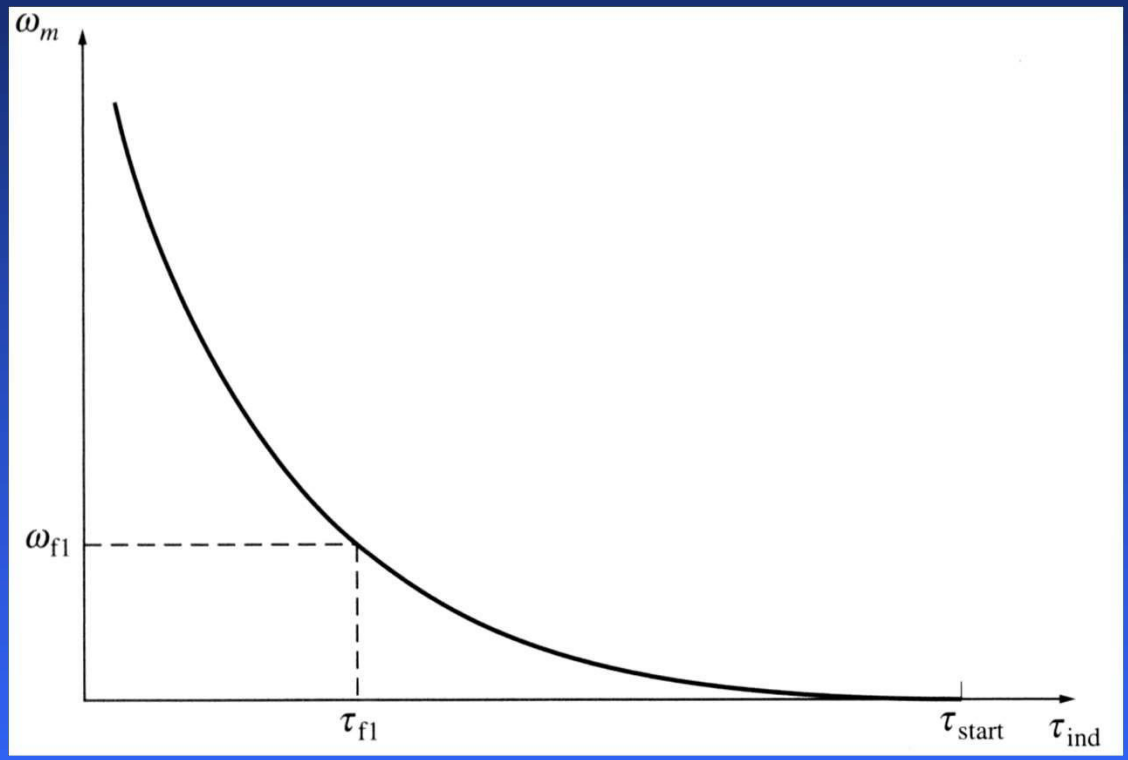




Series motor: terminal characteristic

One serious disadvantage of a series motor is that its speed goes to infinity for a zero torque.

In practice, however, torque never goes to zero because of the mechanical, core, and stray losses. Still, if no other loads are attached, the motor will be running fast enough to cause damage.



Steps must be taken to ensure that a series motor **always** has a load! Therefore, it is not a good idea to connect such motors to loads by a belt or other mechanism that could break.

Series motor: terminal characteristic

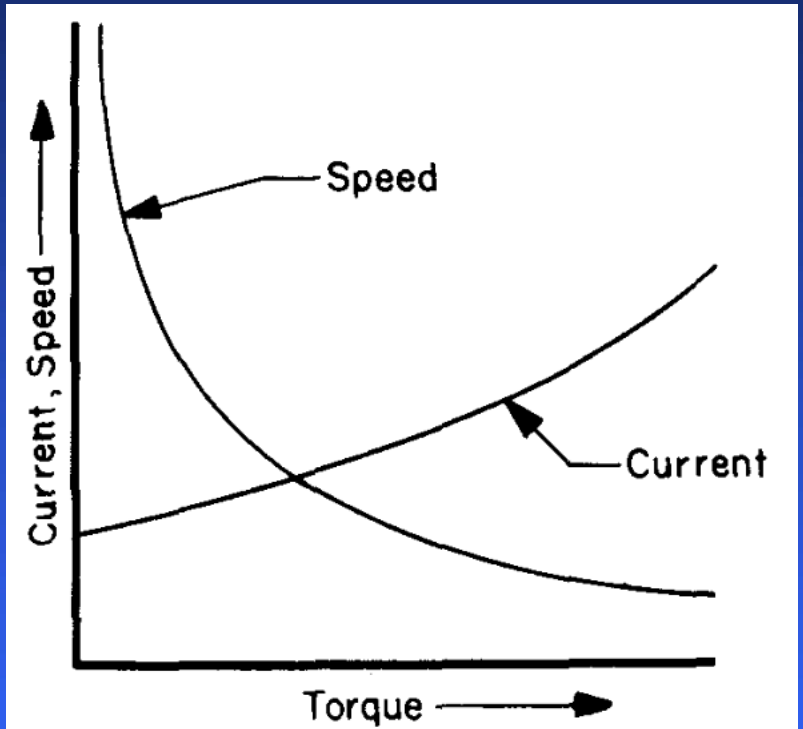


Fig. 2.2.1. Straight-series motor.

$$\tau_{ind} = KcI_A^2$$

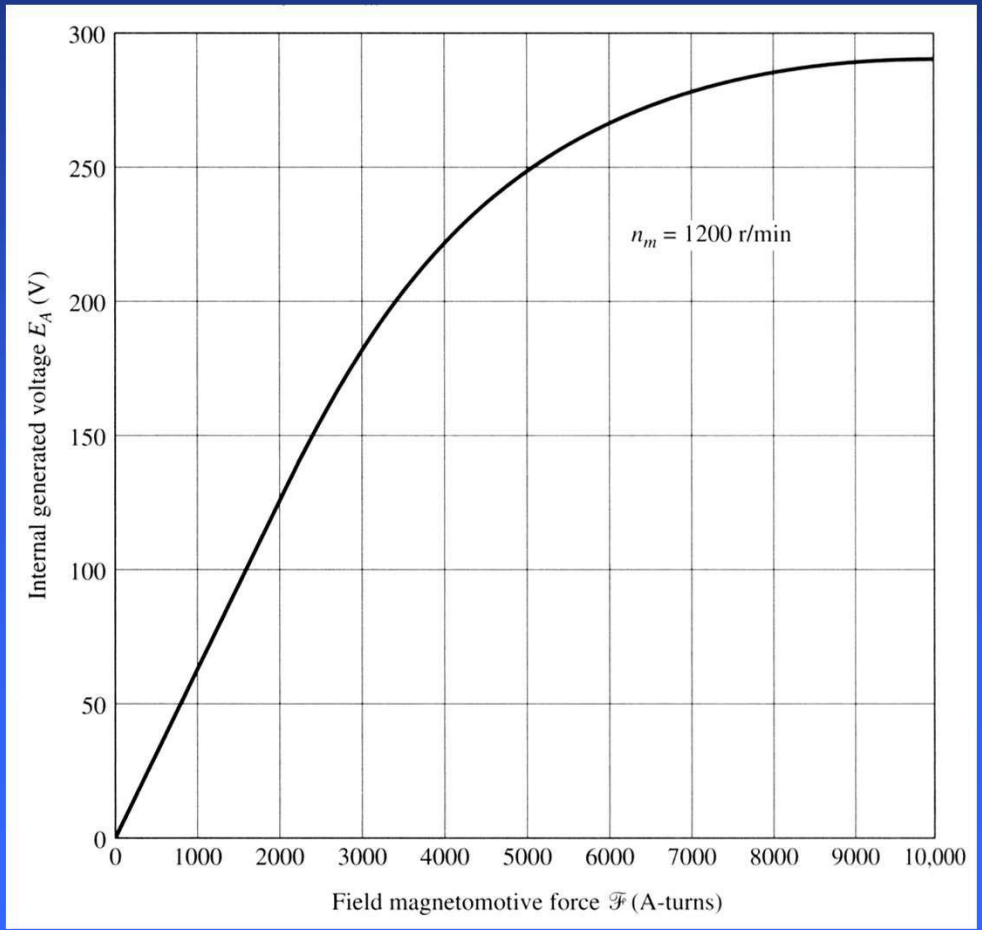
$$\omega = \frac{V_T}{\sqrt{Kc}} \frac{1}{\sqrt{\tau_{ind}}} - \frac{R_A + R_S}{Kc}$$

Series motor: terminal characteristic – Example

Example 5.4: A 250 V series DC motor with compensating windings has a total series resistance $R_A + R_S$ of 0.08 Ω. The series field consists of 25 turns per pole and the magnetization curve is

a) Find the speed and induced torque of this motor when its armature current is 50 A.
b) Calculate and plot its torque-speed characteristic.

a) To analyze the behavior of a series motor with saturation, we pick points along the operating curve and find the torque and speed for each point. Since the magnetization curve is given in units of mmf (ampere-turns) vs E_A for a speed of 1200 rpm, calculated values of E_A must be compared to equivalent values at 1200 rpm.



Series motor: terminal characteristic – Example

For I_A = 50 A

$$E_A = V_T - I_A(R_A + R_S) = 250 - 50 \cdot 0.08 = 246V$$

Since for a series motor $I_A = I_F$ = 50 A, the mmf is

$$\mathrm{F} = NI = 25 \cdot 50 = 1250 \quad A - turns$$

From the magnetization curve, at this mmf, the internal generated voltage is E_{A0} = 80 V. Since the motor has compensating windings, the correct speed of the motor will be

$$n = \frac{E_A}{E_{A0}} n_0 = \frac{246}{80} 1200 = 3690 \quad rpm$$

The resulting torque: $\tau_{ind} = \frac{E_A I_A}{\omega} = \frac{246 \cdot 50}{3690 \cdot 2\pi/60} = 31.8 \quad N\text{-}m$

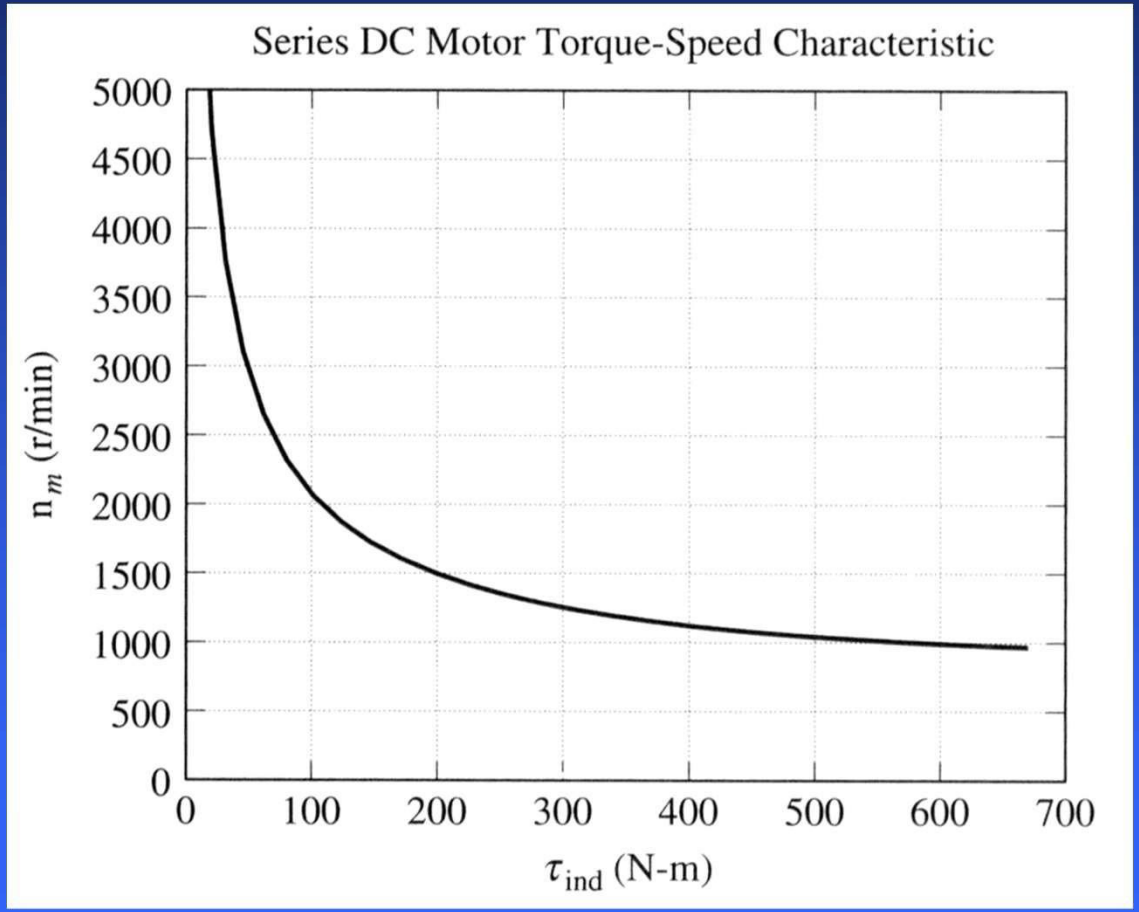




Series motor: terminal characteristic – Example

b) The complete torque-speed characteristic

We notice severe over-speeding at low torque values.



Series motor: Speed control

The only way to control speed of a series DC motor is by changing its terminal voltage, since the motor speed is directly proportional to its terminal voltage ***for any given torque***.

$$\omega = \frac{V_T}{\sqrt{Kc}} \frac{1}{\sqrt{\tau_{ind}}} - \frac{R_A + R_S}{Kc}$$

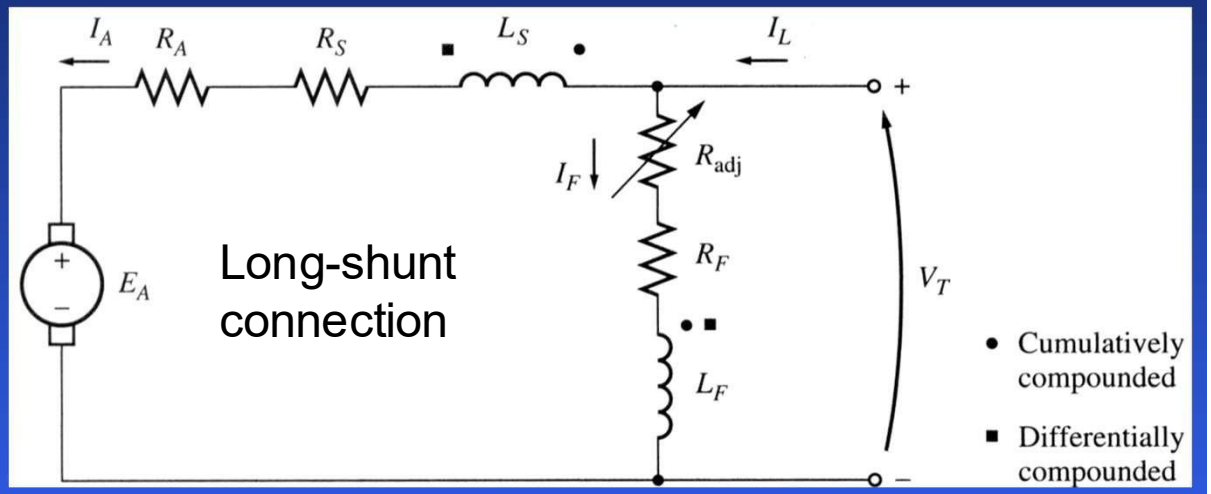
Motor types: Compounded DC motor

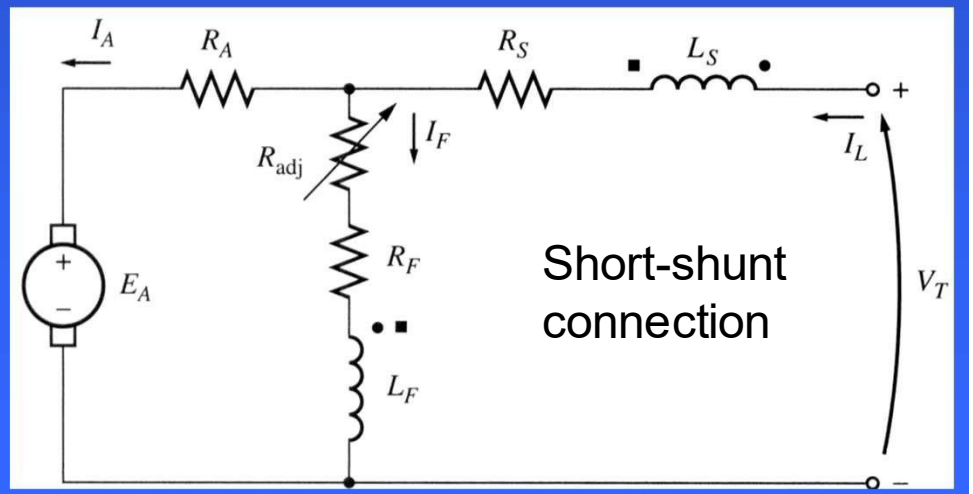
A compounded DC motor is a motor with both a shunt and a series field.

Current flowing into a dotted end of a coil (shunt or series) produces a positive (additive) mmf.

If current flows into the dotted ends of both coils, the resulting mmfs add to produce a larger total mmf – cumulative compounding.

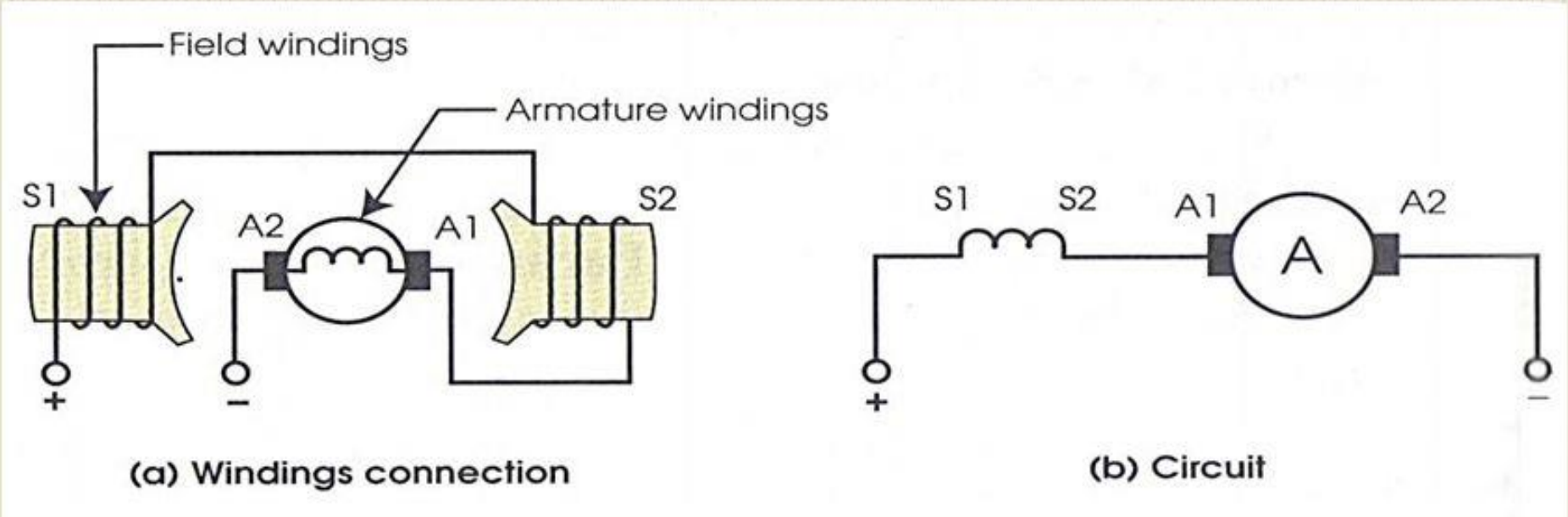
If current flows into the dotted (square) end of one coil and out of the dotted (square) end of another coil, the resulting mmfs is subtractive – differential compounding.





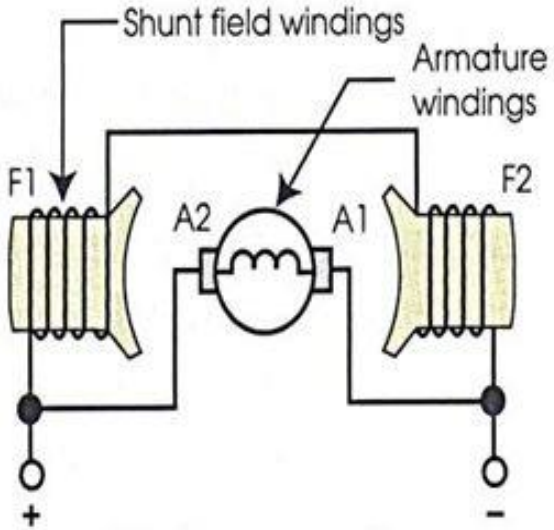
SERIES MOTOR

- In a series motor, the field windings and the armature windings are connected in series to the supply
- The current that flows in the field windings will flow in the armature windings.
- When voltage is supplied, there is no back emf in the armature which enables a high initial current to flow in the field windings and the armature windings.
- This condition generates a high initial torque.
- Thus, the motor is suitable for operating systems with heavy loads

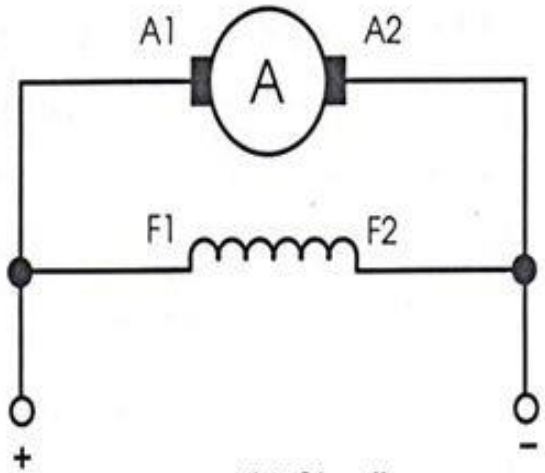


SHUNT MOTOR

- Uses a parallel connection for connecting the field windings and the armature windings to the supply
- The parallel connection causes the current in the field winding to be constant. The current in the armature winding depends on the back emf.
- More suitable when constant speed is required without high initial torque.
- Suitable for use in applications such as machine tools, polishers and sprayers.



(a) Windings connection



(b) Circuit

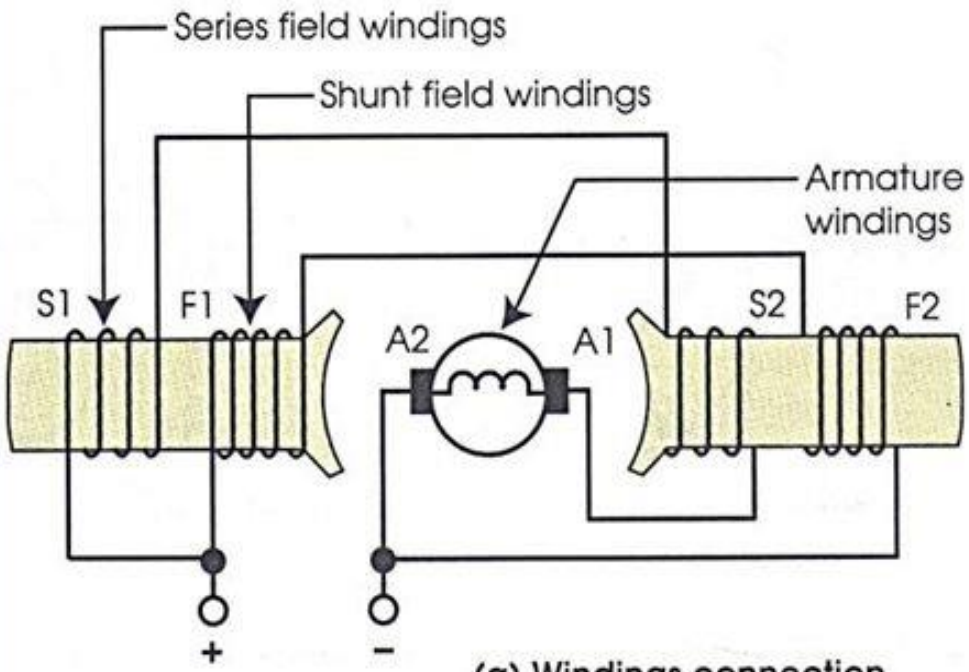
Legend

A1 - A2 : Armature windings

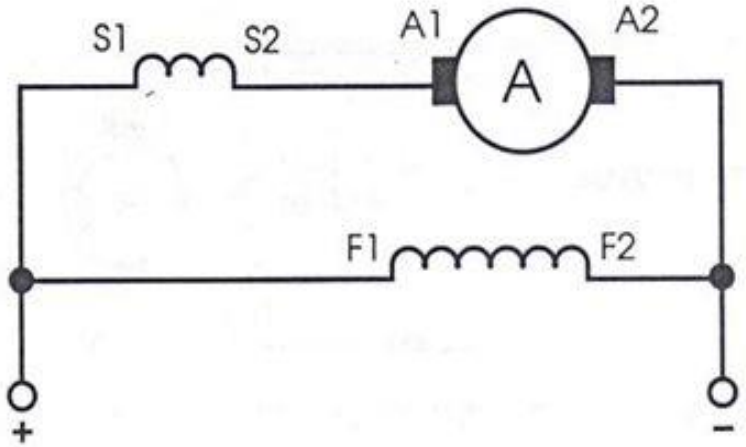
F1 - F2 : Shunt field windings

COMPOUND MOTOR

- The combinations of a series motor and a shunt motor.
- Has the field windings and the armature windings are connected in series to the supply.
- One is connected in series to the armature windings and the power supply. The other is connected in parallel to the supply.
- Has a good starting torque and at constant speed
- Usually used in heavy load machine. Such as in cutting and grinding.



(a) Windings connection



(b) Circuit

Motor types: Compounded DC motor

The Kirchhoff's voltage law equation for a compounded DC motor is

$$V_T = E_A + I_A \left(R_A + R_S \right) \tag{5.85.1}$$

The currents in a compounded DC motor are

$$I_A = I_L - I_F \tag{5.85.2}$$

$$I_F = \frac{V_T}{R_F} \tag{5.85.3}$$

The mmf of a compounded DC motor:

Cumulatively compounded

$$F_{net} = F_F \pm F_{SE} - F_{AR} \tag{5.85.4}$$

Differentially compounded

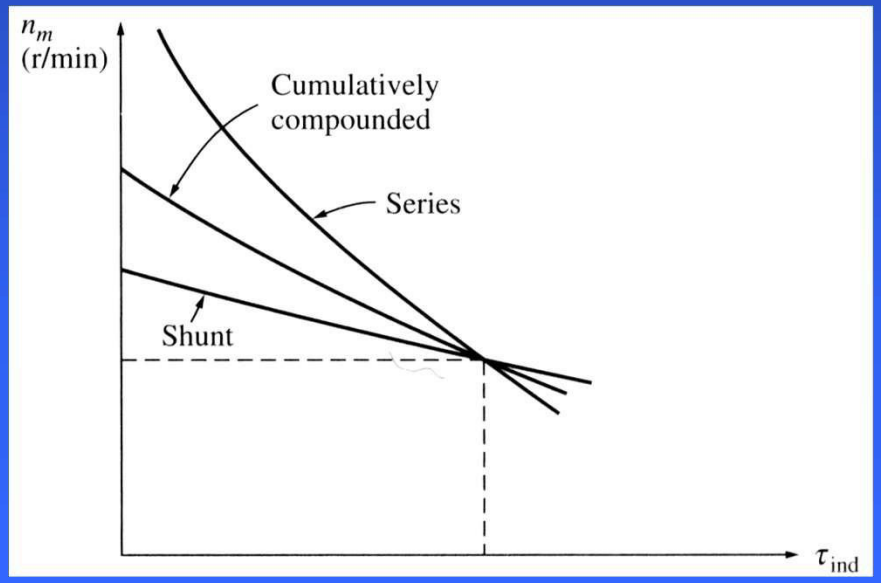
The effective shunt field current in a compounded DC motor:

$$I_F^* = I_F + \frac{N_{SE}}{N_F} I_A - \frac{F_{AR}}{N_F} \tag{5.85.5}$$

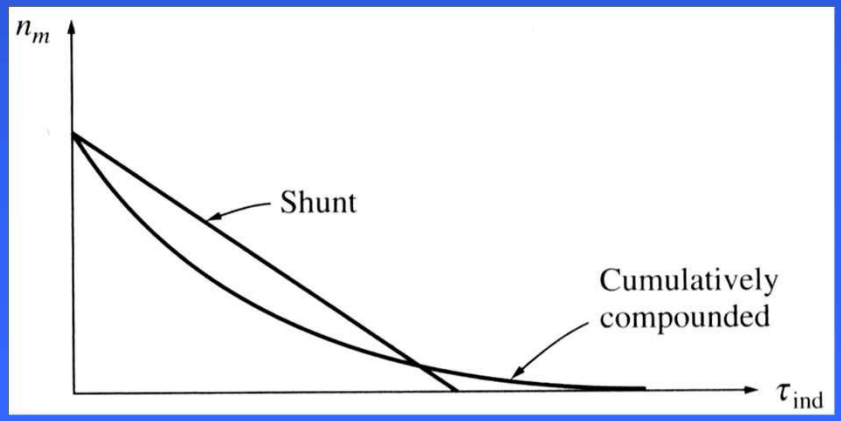
Number of turns

Cumulatively compounded motors: torque-speed characteristic

In a cumulatively compounded motor, there is a constant component of flux and a component proportional to the armature current (and thus to the load). These motors have a higher starting torque than shunt motors (whose flux is constant) but lower than series motors (whose flux is proportional to the armature current).
The series field has a small effect at light loads – the motor behaves as a shunt motor. The series flux becomes quite large at large loads – the motor acts like a series motor.



Similar (to the previously discussed) approach is used for nonlinear analysis of compounded motors.

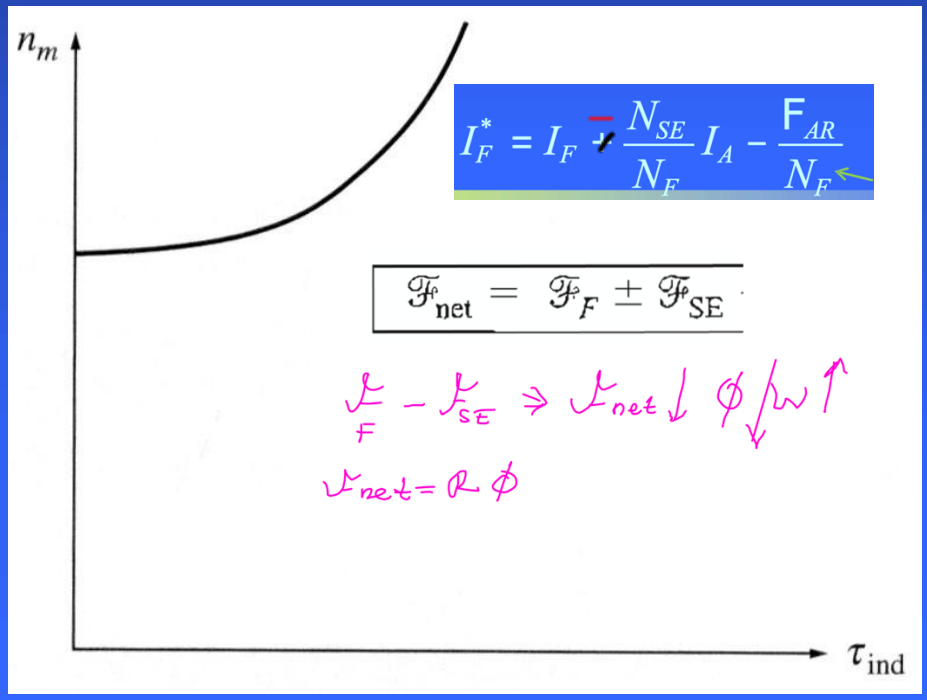


Differentially compounded motors: torque-speed characteristic

Since the shunt mmf and series mmf subtract from each other in a differentially compounded motor, increasing load increases the armature current I_A and decreases the flux. When flux decreases, the motor speed increases further increasing the load. This results in an instability (much worse than one of a shunt motor) making differentially compounded motors unusable for any applications.

In addition to that, these motors are not easy to start… The motor typically remains still or turns very slowly consuming enormously high armature current.

Stability problems and huge starting armature current lead to these motors being never used **intentionally**.

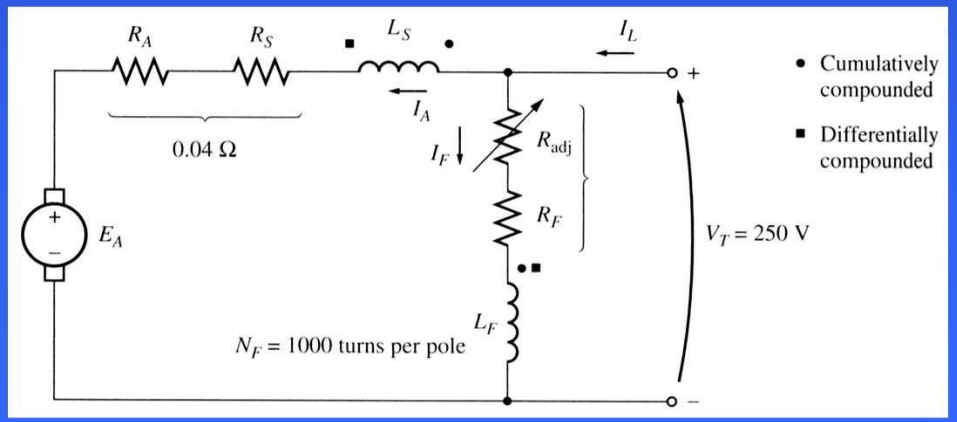


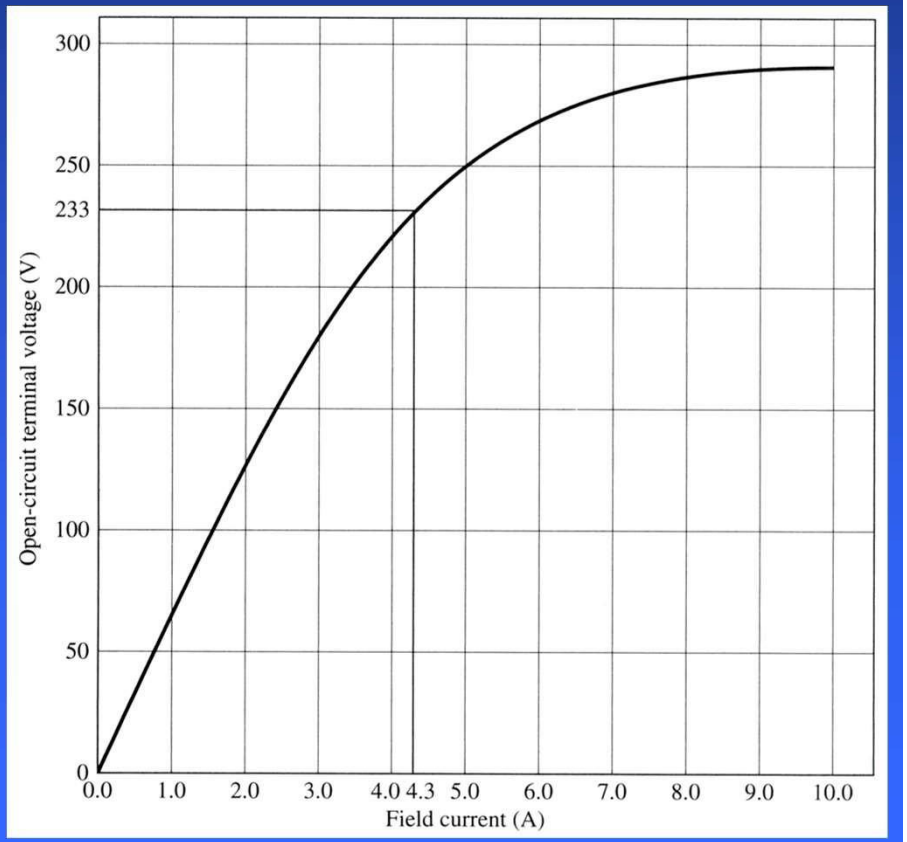
Compounded DC motor: Example

Example 5.5: A 100 hp, 250 V compounded DC motor with compensating windings has an internal resistance, including the series winding of 0.04 Ω. There are 1000 turns per pole on the shunt field and 3 turns per pole on the series windings. The magnetization curve is shown below.

The field resistor has been adjusted for the motor speed of 1200 rpm. The mechanical, core, and stray losses may be neglected.

a) Find the no-load shunt field current.
Find the speed at I_A = 200 A if the motor is b) cumulatively; c) differentially compounded





Compounded DC motor: Example

a) At no load, the armature current is zero; therefore, the internal generated voltage equals V_T = 250 V. From the magnetization curve, a field current of 5 A will produce a voltage E_A = 250 V at 1200 rpm. Therefore, the shunt field current is 5 A.

b) When the armature current is 200 A, the internal generated voltage is

$$E_A = V_T - I_A(R_A + R_S) = 250 - 200 \cdot 0.04 = 242 \ V$$

The effective field current of a cumulatively compounded motor will be

$$I_F^* = I_F + \frac{N_{SE}}{N_F} I_A - \frac{F_{AR}}{N_F} = 5 + \frac{3}{1000} 200 = 5.6 \ A$$

From the magnetization curve, E_{A0} = 262 V at speed n_0 = 1200 rpm. The actual motor speed is

$$n = \frac{E_A}{E_{A0}} n_0 = \frac{242}{262} 1200 = 1108 \ \ rpm$$

Compounded DC motor: Example

c) The effective field current of a differentially compounded motor will be

$$I_F^* = I_F - \frac{N_{SE}}{N_F} I_A - \frac{\mathsf{F}_{AR}}{N_F} = 5 - \frac{3}{1000} 200 = 4.4\ A$$

From the magnetization curve, E_{A0} = 236 V at speed n_0 = 1200 rpm. The actual motor speed is

$$n = \frac{E_A}{E_{A0}} n_0 = \frac{242}{236} 1200 = 1230\ \ rpm$$

Cumulatively compounded motors: speed control

The same two techniques that have been discussed for a shunt motor are also available for speed control of a cumulatively compounded motor.

1. Adjusting the field resistance R_F;
2. Adjusting the armature voltage V_A.

The details of these methods are very similar to already discussed for shunt DC motors.





The flux density of the magnetic field in DC motors depends on the magnetic material properties of the rotor and stator.

The magnetic torque is expressed as a function of the magnetic field and is usually represented by the formula:

$$T = K_t \cdot B_r \cdot B_s \cdot \sin(\theta)$$

where:

- T = Torque produced by the motor.
- K_t = Torque constant, which depends on the motor's construction and winding characteristics.
- B_r = Magnetic flux density of the rotor.
- B_s = Magnetic flux density of the stator.
- θ = Angle between the rotor and stator magnetic fields.
- The stronger the rotor and stator magnetic fields, the greater the torque.
- The torque fluctuates depending on the angle θ, reaching maximum when $\theta = 90^\circ$, i.e., when the rotor and stator magnetic fields are orthogonal.

DC motor starters

In order for DC motors to function properly, they must have some special control and protection equipment associated with them. The purposes of this equipment are:

1. To protect the motor against damage due to short circuits in the equipment;
2. To protect the motor against damage from long-term overloads;
3. To protect the motor against damage from excessive starting currents;
4. To provide a convenient manner in which to control the operating speed of the motor.



DC motor problems on starting

At starting conditions, the motor is not turning, therefore the internal generated voltage E_A = 0V. Since the internal resistance of a normal DC motor is very low (3-6 % pu), a very high current flows.

For instance, for a 50 hp, 250 V DC motor with armature resistance R_A of 0.06 Ω and a full-load current about 200 A, the starting current is

$$I_A = \frac{V_T - E_A}{R_A} = \frac{250 - 0}{0.06} = 4167\ A$$

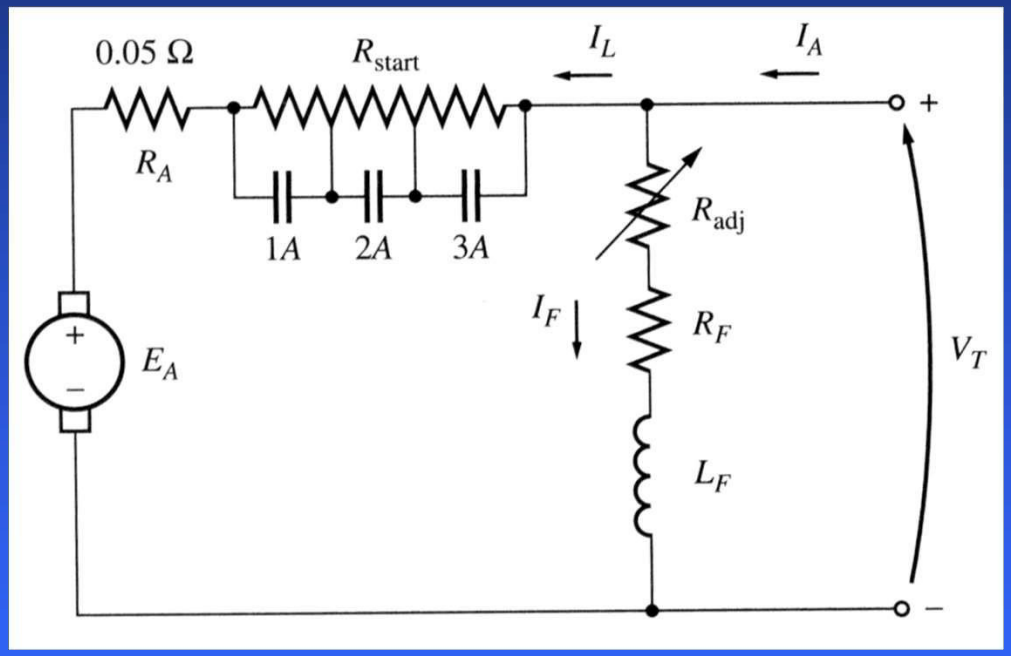
This current is over 20 times the motor's rated full-load current and may severely damage the motor.

A solution to the problem of excessive starting current is to insert a starting resistor in series with the armature to limit the current until E_A can build up to limit the armature current. However, this resistor must be removed from the circuit as the motor speed is high since otherwise such resistor would cause losses and would decrease the motor's torque-speed characteristic.

DC motor problems on starting

In practice, a starting resistor is made up of a series of resistors that can be successively removed from the circuit as the motor speeds up.

A shunt motor with an extra starting resistor that can be cut out of the circuit in segments by closing the 1A, 2A, and 3A contacts.

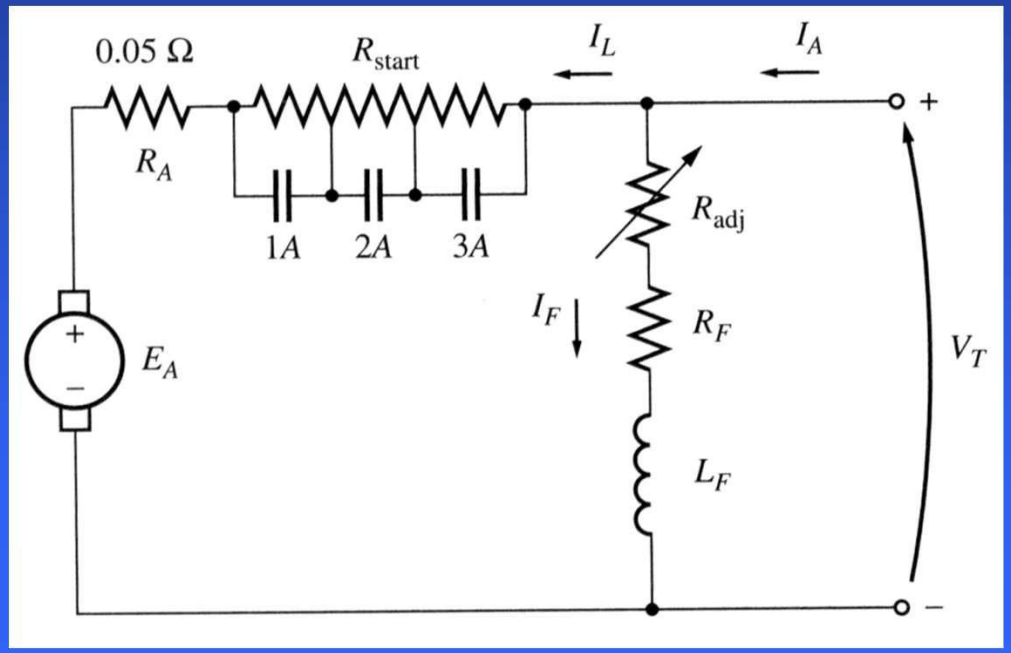


Therefore, two considerations are needed to be taken into account: Select the values and the number of resistor segments needed to limit the starting current to desired ranges; Design a control circuit shutting the resistor bypass contacts at the proper time to remove particular parts of the resistor from the circuit.

DC motor problems on starting: Ex

Example 5.6: A 100 hp, 250 V 350 A shunt DC motor with an armature resistance of 0.05 Ω needs a starter circuit that will limit the max starting current to twice its rated value and which will switch out sections of resistor once the armature current decreases to its rated value.

a. How many stages of starting resistance will be required to limit the current to the specified range?
b. What must the value of each segment of the resistor to be? At what voltage should each stage of the starting resistance be cut out?



DC motor problems on starting: Ex

a. The starting resistor must be selected such that the current flow at the start equals twice the rated current. As the motor speeds up, an internal voltage E_A (which opposes the terminal voltage of the motor and, therefore, limits the current) is generated. When the current falls to the rated value, a section of the starting resistor needs to be taken out to increase the current twice. This process (of taking out sections of the starting resistor) repeats until the entire starting resistance is removed. At this point, the motor's armature resistance will limit the current to safe values by itself.

The original resistance in the starting circuit is

$$R_{tot} = R_1 + R_2 + \ldots + R_A = \frac{V_T}{I_{\max}}$$

After the stages 1 through *i* are shorted out, the total resistance left in the starting circuit is

$$R_{tot,i} = R_{i+1} + \ldots + R_A$$

DC motor problems on starting: Ex

The resistance R_1 must be switched out of the circuit when the armature current falls to

$$I_{A,\min} = \frac{V_T - E_{A,1}}{R_{tot}} = I_{\min} = 350\ A$$

After the resistance R_1 is out of the circuit, the armature current must increase to

$$I_{A,\max} = \frac{V_T - E_{A,2}}{R_{tot,1}} = I_{\max} = 700\ A$$

Since $E_A = K\phi\omega$ the quantity $V_T - E_A$ must be constant when the resistance is switched out. Therefore

$$I_{\min} R_{tot} = V_T - E_A = I_{\max} R_{tot,1}$$

The resistance left in the circuit is

$$R_{tot,1} = \frac{I_{\min}}{I_{\max}} R_{tot} \quad \Rightarrow \quad R_{tot,n} = \left(\frac{I_{\min}}{I_{\max}} \right)^n R_{tot}$$

DC motor problems on starting: Ex

The starting process is completed when $R_{tot,n}$ is not greater than the internal armature resistance R_A. At the boundary:

$$R_A = R_{tot,n} = \left(\frac{I_{\min}}{I_{\max}} \right)^n R_{tot}$$

Solving for n:

$$n = \frac{\log\left(R_A / R_{tot}\right)}{\log\left(I_{\min} / I_{\max}\right)}$$

Notice that the number of stages n must be rounded up to the next integer.

$$R_{tot} = \frac{V_T}{I_{\max}} = \frac{250}{700} = 0.357\,\Omega$$

$$n = \frac{\log\left(R_A / R_{tot}\right)}{\log\left(I_{\min} / I_{\max}\right)} = \frac{\log\left(0.05/0.357\right)}{\log\left(350/700\right)} = 2.84 \approx 3$$





DC motor problems on starting: Ex

b. The armature circuit will contain the armature resistance R_A and three starting resistors. At first, E_A = 0, I_A = 700 A, and the total resistance is 0.357 Ω. The total resistance will be in the circuit until the current drops to 350 A. This occurs when

$$E_{A,1} = V_T - I_{A,\min} R_{tot} = 250 - 350 \cdot 0.357 = 125V$$

At this time, the starting resistor R_1 will be taken out making

$$R_{tot,1} = R_A + R_2 + R_3 = \frac{V_T - E_{A,1}}{I_{\max}} = \frac{250 - 125}{700} = 0.1786\,\Omega$$

This (new) total resistance will be in the circuit until the current drops again to 350 A. This occurs when

$$E_{A,2} = V_T - I_{A,\min} R_{tot,1} = 250 - 350 \cdot 0.1786 = 187.5V$$

At this time, the starting resistor R_2 will be taken out leaving

$$R_{tot,2} = R_A + R_3 = \frac{V_T - E_{A,2}}{I_{\max}} = \frac{250 - 187.5}{700} = 0.0893\,\Omega$$





DC motor problems on starting: Ex

This total resistance will be in the circuit until the current drops again to 350 A. This occurs when

$$E_{A,3} = V_T - I_{A,\min} R_{tot,2} = 250 - 350 \cdot 0.0893 = 218.75 V$$

At this time, the starting resistor R_3 will be taken out leaving only R_A in the circuit. The motor's current at that moment will increase to

$$I_{A,3} = \frac{V_T - E_{A,3}}{R_A} = \frac{250 - 218.75}{0.05} = 625\ A$$

which is less than the allowed value. Therefore, the resistances are

$$R_3 = R_{tot,3} - R_A = 0.0893 - 0.05 = 0.0393\ \Omega$$

$$R_2 = R_{tot,2} - R_3 - R_A = 0.1786 - 0.0393 - 0.05 = 0.0893\ \Omega$$

$$R_1 = R_{tot,1} - R_2 - R_3 - R_A = 0.357 - 0.1786 - 0.0393 - 0.05 = 0.1786\ \Omega$$

The resistors R_1, R_2, and R_3 are cut out when E_A reaches 125 V, 187.5 V, and 218.75 V, respectively.



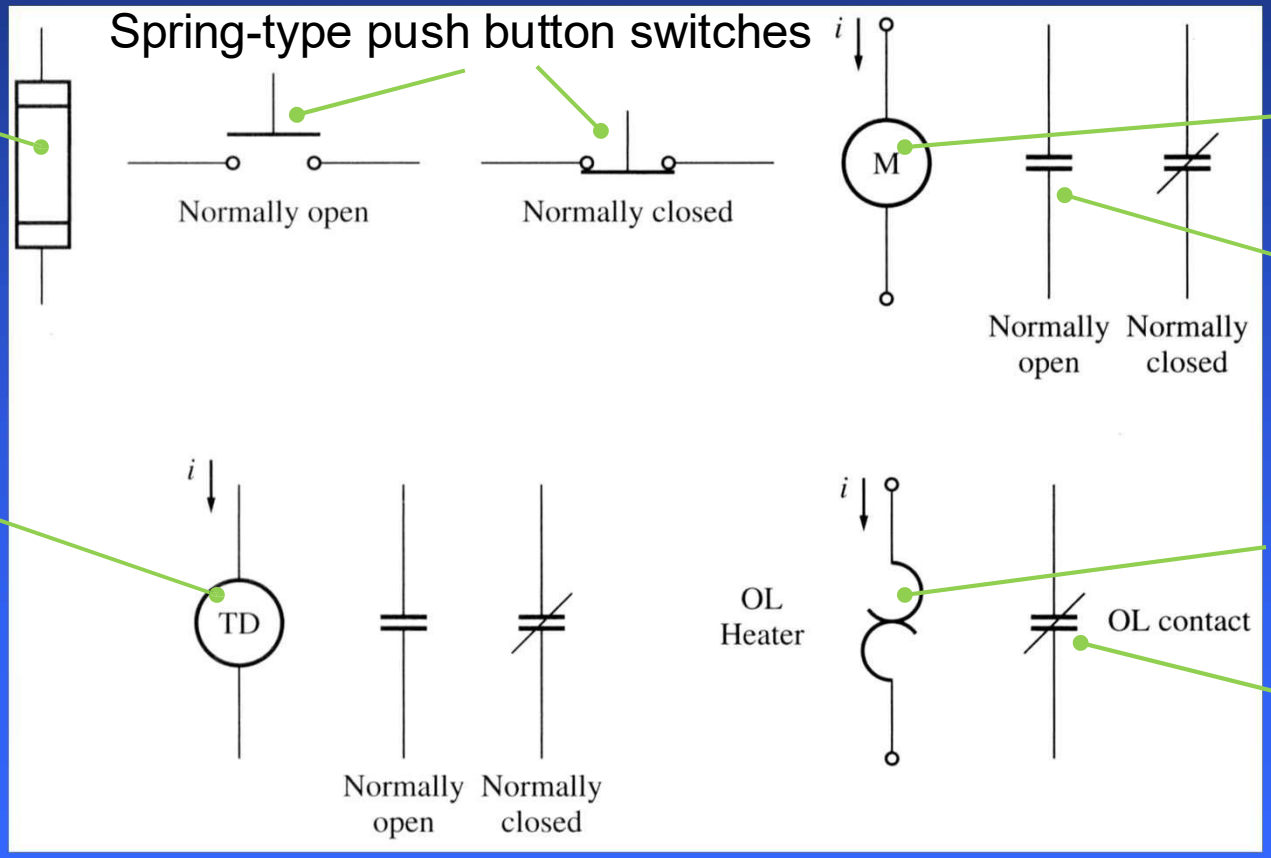


DC motor starting circuits

Several different schemes can be used to short contacts and cut out the sections of a starting resistor. Some devices commonly used in motor-control circuits are

Fuses: protects against short circuits

Time delay relay similar to ordinary relay except for having adjustable time delay.



Relay: a main coil and a number of contacts

Overload: a heater coil and normally closed contacts

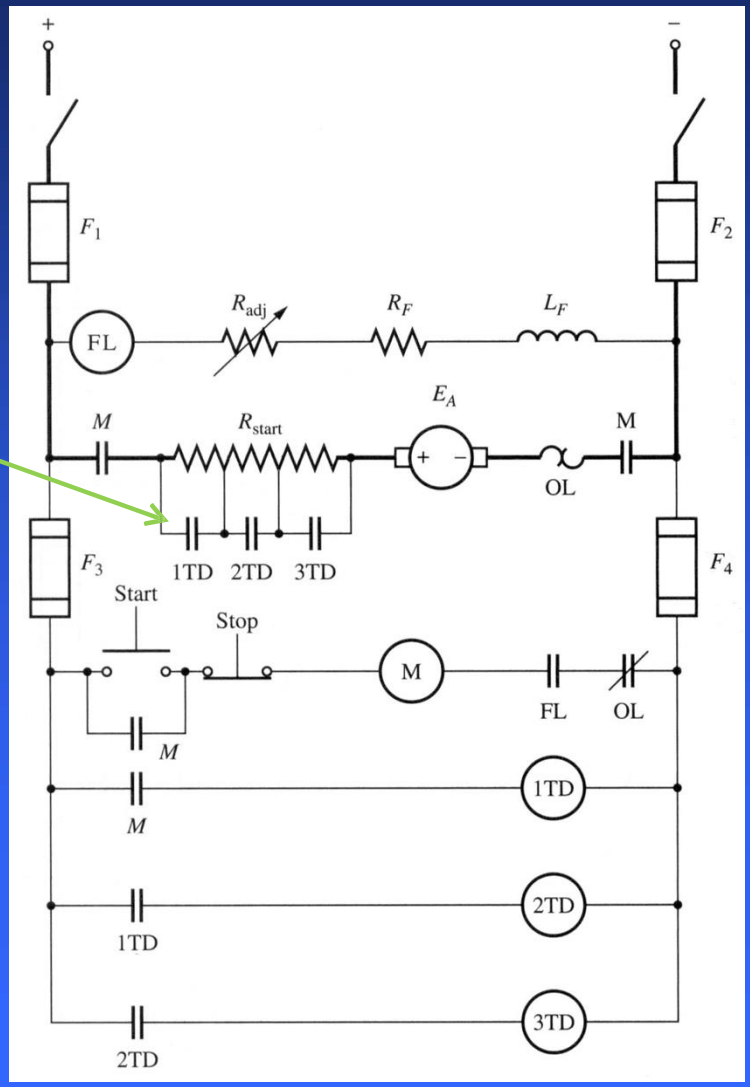
DC motor starting circuits

A common DC motor starting circuit:

A series of time delay relays shut contacts removing each section of the starting resistor at approximately correct times.

Notice that the relay 1TD is energized at the same time as the motor starts – contacts of 1TD will shut a part of the starting resistor after some time. At the same instance, relay 2TD is energized and so on…

Observe also 4 fuses protecting different parts of the circuit and the overload in series with the armature winding.



DC motor starting circuits

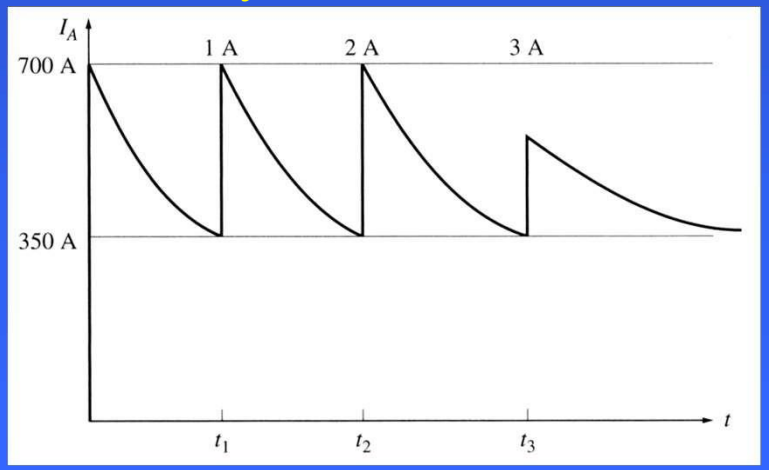
Another type of motor starter:

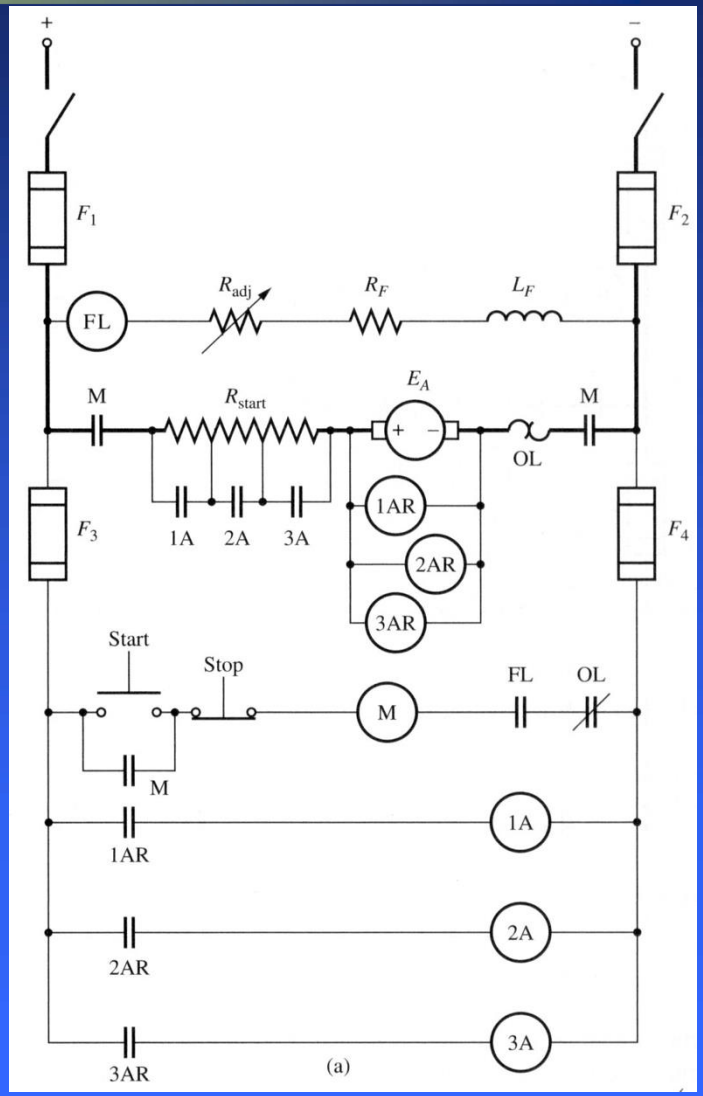
A series of relays sense the value of armature voltage E_A and cut out the starting resistors as it riches certain values.

This starter type is more robust to different loads.

FL is the *field loss relay*: if the field is lost for any reason, power to the M relay will be turned off.

Armature current in a DC motor during starting.





DC motor efficiency calculations

To estimate the efficiency of a DC motor, the following losses must be determined:

1. Copper losses;
2. Brush drop losses;
3. Mechanical losses;
4. Core losses;
5. Stray losses.

To find the copper losses, we need to know the currents in the motor and two resistances. In practice, the armature resistance can be found by blocking the rotor and a small DC voltage to the armature terminals: such that the armature current will equal to its rated value. The ratio of the applied voltage to the armature current is approximately R_A.

The field resistance is determined by supplying the full-rated field voltage to the field circuit and measuring the resulting field current. The field voltage to field current ratio equals to the field resistance.





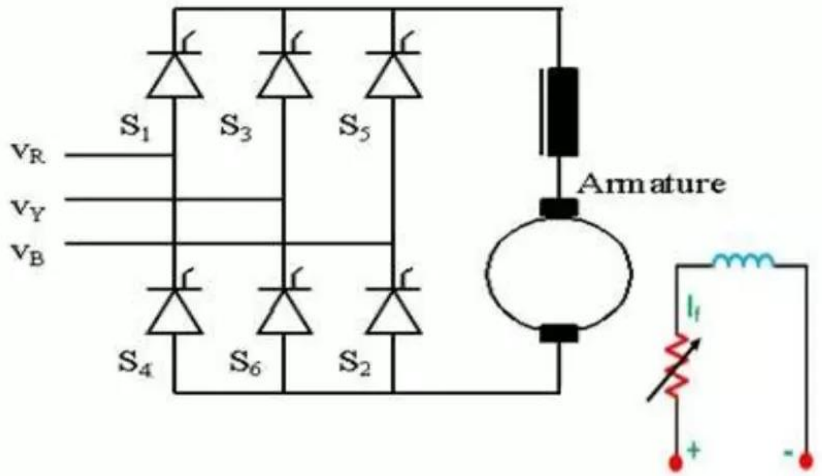
DC motor efficiency calculations

Brush drop losses are frequently lumped together with copper losses. If treated separately, brush drop losses are a product of the brush voltage drop V_{BD} and the armature current I_A.

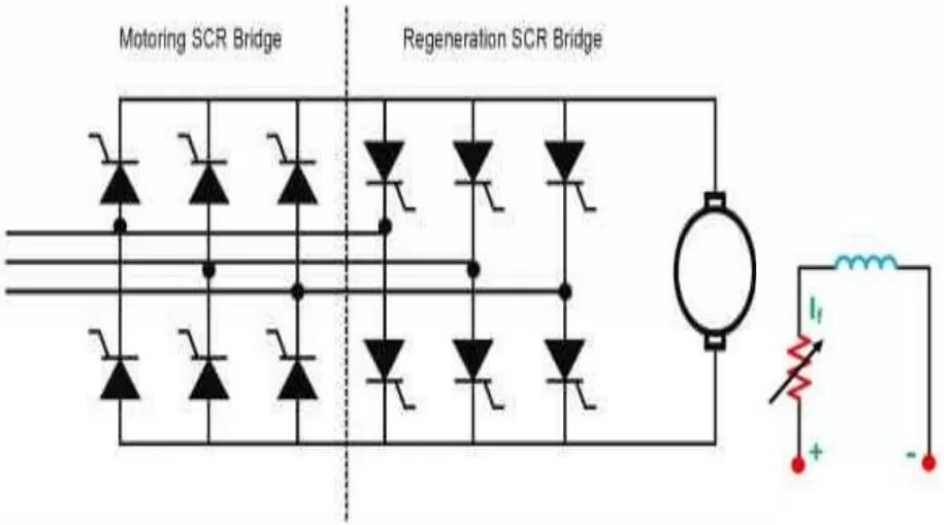
The core and mechanical losses are usually determined together. If a motor is running freely at no load and at the rated speed, the current I_A is very small and the armature copper losses are negligible. Therefore, if the field copper losses are subtracted from the input power of the motor, the remainder will be the mechanical and core losses. These two losses are also called the no-load rotational losses. As long as the motor's speed remains approximately the same, the no-load rotational losses are a good estimate of mechanical and core losses in the machine under load.







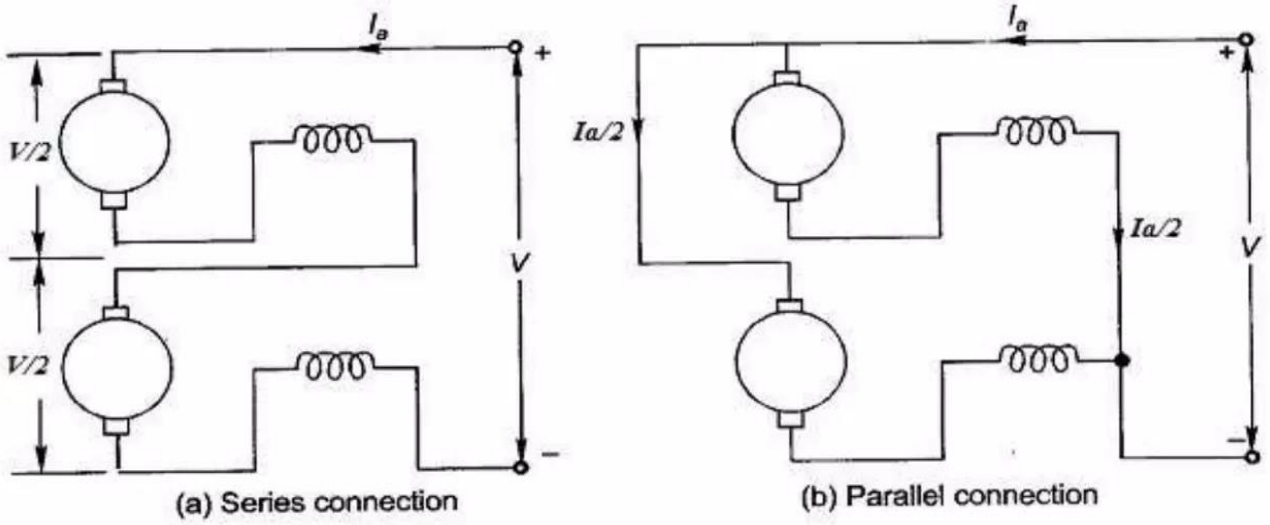
(a) A Two quadrant solid state DC motor controller



(b) A Four quadrant solid state DC motor controller

3. Series-parallel armature Control:

- It requires two identical DC motors coupled together mechanically to a common load.
- Commonly employ for DC series motors.
- Application: DC series traction motors



Series-parallel speed control of two dc series motors

- In parallel the speed is four times than the speed in series connection

Application Areas of Types of Motors

Type of Motor	Characteristic	Application
Shunt	Approximately constant speed, which is adjustable, but medium stating torque.	Machine tools, lathes, positive displacement pumps and blowers, compressors, reciprocating pumps etc. In fact, wherever constant speed is desirable.
Series	Variable speed, which is adjustable. High starting torque.	Traction work i.e. electric locomotives, rapid transit systems, trams etc. Crane hoists, welder machines, conveyors etc.
Cumulative Compound	Variable speed, which is adjustable. High starting torque.	Intermittent high torque loads, shears and punches, elevators, conveyors, heavy planers, rolling mills, crushers etc.